

Model 1918-R

Hand-held Optical Meter



User's Manual



Newport®
Experience | Solutions

EU Declaration of Conformity



We declare that the accompanying product, identified with the **CE** mark, complies with requirements of the Electromagnetic Compatibility Directive, 2004/108/EC and the Low Voltage Directive 73/23/EEC.

Model Number: 1918-R

Year **CE mark affixed:** 2007

Type of Equipment: Electrical equipment for measurement, control and laboratory use.

Manufacturer: Newport Corporation

1791 Deere Avenue

Irvine, California 92606

Standards Applied:

Compliance was demonstrated to the following standards to the extent applicable:

BS EN61326-1: 1997+A1+A2 +A3 “Electrical equipment for measurement, control and laboratory use – EMC requirements”

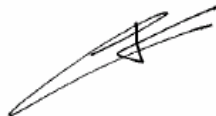
Performance criteria B was achieved during immunity tests.

This equipment meets the CISPR 11:2006+A2 Class A Group 1 radiated and conducted emission limits.

BS EN 61000-3-2:2001, Harmonic current emissions, Class A

BS EN 61000-3-3:2002, Voltage fluctuations and flicker

BS EN 61010-1:2001, 2nd Edition “Safety requirements for electrical equipment for measurement, control and laboratory use”



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Firmware and User Manual Updates

Dear Customer,

In an effort to keep the 1918-R Optical Meter optimized for your applications, Newport will on occasion update existing, and add new features to this instrument.

To utilize this new functionality will require an update to the instrument's firmware, which can be easily accomplished by the user, as described in this User Manual. As required, Newport will also generate a new version of this User Manual, reflecting updates to the instrument.

Please check the Newport website (www.Newport.com) for newer versions of the firmware and the User Manual, which can be downloaded as a PDF file. Call your local Newport application specialist if you need support with locating or downloading these files.

Enjoy your new instrument!

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Newport Corporation warrants that this product will be free from defects in material and workmanship and will comply with Newport's published specifications at the time of sale for a period of one year from date of shipment. If found to be defective during the warranty period, the product will either be repaired or replaced at Newport's option.

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1791 Deere Avenue
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Part No. 90039766, Rev. A

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Newport Corporation Calling Procedure

If there are any defects in material or workmanship or a failure to meet specifications, promptly notify Newport's Returns Department by calling 1-800-222-6440 or by visiting our website at www.newport.com/returns within the warranty period to obtain a **Return Material Authorization Number (RMA#)**. Return the product to Newport Corporation, freight prepaid, clearly marked with the RMA# and we will either repair or replace it at our discretion. Newport is not responsible for damage occurring in transit and is not obligated to accept products returned without an RMA#.

E-mail: rma.service@newport.com

When calling Newport Corporation, please provide the customer care representative with the following information:

- Your Contact Information
- Serial number or original order number
- Description of problem (i.e., hardware or software)

To help our Technical Support Representatives diagnose your problem, please note the following conditions:

- Is the system used for manufacturing or research and development?
- What was the state of the system right before the problem?
- Have you seen this problem before? If so, how often?
- Can the system continue to operate with this problem? Or is the system non-operational?
- Can you identify anything that was different before this problem occurred?

IMPORTANT NOTE

Before plugging the instrument into a PC via a USB communication port, please make sure that the USB Drivers are installed. Run Setup.exe from the Software CD that came with your product. The installation program will configure the PC with the 1918-R USB drivers.

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1 Safety Precautions

1.1 Definitions and Symbols

The following terms and symbols are used in this documentation and also appear on the 1918-R Optical Meter where safety-related issues occur.

1.1.1 General Warning or Caution



Figure 1 General Warning or Caution Symbol

The Exclamation Symbol in the figure above appears in Warning and Caution tables throughout this document. This symbol designates an area where personal injury or damage to the equipment is possible.

1.1.2 Electric Shock



Figure 2 Electrical Shock Symbol

The Electrical Shock Symbol in the figure above appears throughout this manual. This symbol indicates a hazard arising from dangerous voltage. Any mishandling could result in irreparable damage to the equipment, and personal injury or death.

1.1.3 TÜV Mark



Figure 3 TÜV Mark

The presence of the TÜV mark indicates that it has been designed, tested and certified as complying with the safety requirements in conjunction with the issuance of this mark.

1.1.4 European Union CE Mark



Figure 4 *CE Mark*

The presence of the CE Mark on Newport Corporation equipment means that it has been designed, tested and certified as complying with all applicable European Union (CE) regulations and recommendations.

1.1.5 Alternating voltage symbol



Figure 5 *Alternating Voltage Symbol*

This international symbol implies an alternating voltage or current.

1.1.6 Stand-by



Figure 6 *Stand-by Symbol*

The Stand-by Symbol in the figure above represents a power switch position. This symbol represents a Stand-by condition in which some circuitry is still on. In the case of the 1918-R Optical Power Meter, in the Stand-by position the charging circuit is still on, while the main processing circuit is off.

1.1.7 Frame or Chassis

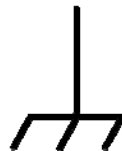


Figure 7 *Frame or Chassis Terminal Symbol*

The symbol in the figure above appears on the 1918-R Optical Meter. This symbol identifies the frame or chassis terminal

1.1.8 Waste Electrical and Electronic Equipment (WEEE)



Figure 8 WEEE Directive Symbol

This symbol on the product or on its packaging indicates that this product must not be disposed of with regular waste. Instead, it is the user responsibility to dispose of waste equipment according to the local laws. The separate collection and recycling of the waste equipment at the time of disposal will help to conserve natural resources and ensure that it is recycled in a manner that protects human health and the environment. For information about where the user can drop off the waste equipment for recycling, please contact your local Newport Corporation representative. See Section 16 for instructions on how to disassemble the equipment for recycling purposes.

1.1.9 Control of Hazardous Substances

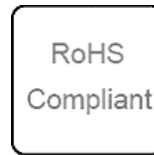


Figure 9 RoHS Compliant Symbol

This label indicates the products comply with the EU Directive 2002/95/EC that restricts the content of six hazardous chemicals.

1.2 Warnings and Cautions

The following are definitions of the Warnings, Cautions and Notes that are used throughout this manual to call your attention to important information regarding your safety, the safety and preservation of your equipment or an important tip.

	WARNING Situation has the potential to cause bodily harm or death.
---	--

**CAUTION**

Situation has the potential to cause damage to property or equipment.

NOTE

Additional information the user or operator should consider.

1.2.1**General Warnings**

Observe these general warnings when operating or servicing this equipment:

- Heed all warnings on the unit and in the operating instructions.
- Do not use this equipment in or near water.
- Route power cords and other cables so they are not likely to be damaged.
- Disconnect power before cleaning the equipment. Do not use liquid or aerosol cleaners; use only a damp lint-free cloth.
- Lockout all electrical power sources before servicing the equipment.
- To avoid fire hazard, use only the specified fuse(s) with the correct type number, voltage and current ratings as referenced in the appropriate locations in the service instructions or on the equipment. Only qualified service personnel should replace fuses.
- To avoid explosion, do not operate this equipment in an explosive atmosphere.
- Qualified service personnel should perform safety checks after any service.

1.2.2 General Cautions

Observe these cautions when operating or servicing this equipment:

- If this equipment is used in a manner not specified in this manual, the protection provided by this equipment may be impaired.
- Do not position this product in such a manner that would make it difficult to disconnect the power cord.
- Use only the specified replacement parts.
- Follow precautions for static sensitive devices when handling this equipment.
- This product should only be powered as described in the manual.
- There are no user-serviceable parts inside the 1918-R Optical Meter with the exception of the battery.
- Use only the Newport Model 1918-BAT battery.
- Use only the Newport Model 1918-PS external power supply.

- To prevent damage to the equipment, read the instructions in the equipment manual for proper input voltage.
- Adhere to good laser safety practices when using this equipment.

1.3 Location of Labels

1.3.1 Rear Panel

Markings on the rear panel identify the instrument compliance with different standards and regulations. The labels are located in the middle of the rear panel.

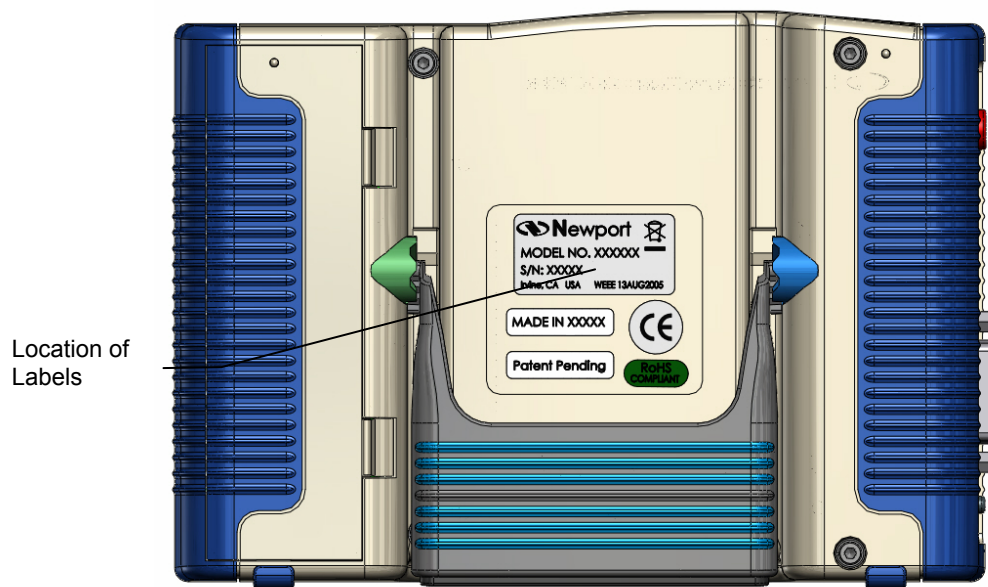


Figure 10 Locations of labels on the rear panel

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2 General Information

2.1 Introduction

The 1918-R Optical Meter is designed to provide a powerful combination of features to measure optical power and energy of near-monochromatic or monochromatic sources. The instrument is powered by internal, rechargeable batteries, or via the AC wall-plug, hence making it a useful tool for both stationary and portable applications.

Use of the extensive measurement modes and features have been simplified with an intuitive menu driven structure that provides quick access to all modes, features and settings. Some of the prominent specifications and features that will simplify your calibrated measurement applications are in the following:

- Compatibility with Photodiode, Thermopile and Pyroelectric detectors. (Low-Power (Semiconductor) Family, High-Power (Thermopile) Family, Energy (Pyroelectric) Family, PMT, GaN, PbS, PbSe, and HgCdZnTe)
- Measurement rate up to 4 kHz with internal signal sampling rate of 250 kHz.
- Multiple measurement modes for power and energy measurements: Single, Continuous, Integrated, and Peak-to-Peak .
- Software suite, including LabVIEW drivers and Windows application.
- 250,000 internal data point storage.
- External USB flash-memory compatibility.
- USB Device interface.

The full color TFT 4" LCD renders excellent visibility both with the naked eye and laser goggles for single screen rendering of plotted and enumerate results. Measurements can be displayed in W, Joules, Amps, Volts, dBm, dB, Sun or relative units, either directly or as relative ratio measurements from present or stored values. Statistical capabilities include the computation of Min, Max, Max-Min, Mean and Standard Deviation. Additional features such as digital and analog filtering, and data storage of up to 250,000 readings per channel are also offered.

Newport's experience with calibration, together with N.I.S.T. calibration traceability and high precision Optical Meters provide users with accurate measurements and exceptional inter-instrument correlation. In R&D, QA/QC,

and manufacturing environments, the 1918-R Series optical meters enable users to benefit from high correlation between multiple locations at a price-to-performance ratio second to none. Among all the other practical tools provided, remote controlling with a computer and synchronization to other instruments are simplified with the inclusion of LabVIEW drivers, tools to develop in the .NET environment and a high-speed software utility that fully utilized the optical meter's ability to sample at 250 kHz and transfer data via a Full-Speed USB interface (11 Mbps).

2.2 Calibration

Calibration of the optical meter is done at the factory by defining a slope and offset, for all ranges as determined for each detector type.

Newport recommends annual factory re-calibration to ensure the continued accuracy of optical meter measurements.

Please refer to the "Maintenance and Troubleshooting" section for contact information for re-calibration of your optical meter.

2.3 Specifications

2.3.1 Physical Specifications:

Dimensions:	5.4 x 7.6 x 2.4 inch (137 x 193 x 61 mm)
Weight:	max. 2.3 lb
Enclosure:	Plastic molded case with a kick stand.
Connectors:	Optical Detector 15-Pin D-Sub, Analog Output 3.5 mm Jack, USB-A Host, mini-B USB Device, DC Input 2.1 mm Power Jack.
Power:	90-264 VAC, 50/60 Hz, 33 Watts with 1918-EXT-PS External Power Supply
Display:	Graphical LCD ¼ VGA, 4 inch diagonal
Display Update Rate:	up to 20 Hz for selected graphical modes
Operating Environment ¹ :	5°C to 40°C; < 70% RH non-condensing
Storage Environment:	-20°C to 60°C; < 90% RH non-condensing
Altitude	< 3000m
Installation Category	II
Pollution Degree	2
Use Location	Indoor use only.

¹ Charging battery up to 35°C

2.3.2 Electrical Specifications

DC Current Measurement (Low-Power, Semiconductor Photodiode)

Signal Range ^{2,3}	0	1	2	3	4
Full-Scale Current	2.50 μ A	25.0 μ A	250 μ A	2.50 mA	25 mA
Resolution (Filtered) ¹	10 pA	100 pA	1.00 nA	10.0 nA	100 nA
Bandwidth ⁶ (Unfiltered)	480 Hz	4 kHz	43 kHz	170 kHz	500 kHz
Accuracy (Filtered) ¹	0.2%	0.2%	0.2%	0.2%	0.2%
Accuracy (Maximum Measurement Rate=4 kHz)	1.0%	1.0%	1.0%	1.0%	1.0%
Signal to Noise Ratio ¹	100 dB	100 dB	100 dB	100 dB	100 dB

Peak-Peak Current Measurement (Low-Power, Semiconductor Photodiode)

Signal Range ^{2,3}	0	1	2	3	4
Full-Scale Current	2.50 μ A	25.0 μ A	250 μ A	2.50 mA	25 mA
Resolution	76.3 pA	763 pA	7.63 nA	76.3 nA	763 nA
Accuracy	1.0%	1.0%	1.0%	1.0%	1.0%
Bandwidth ⁶	480 Hz	4 kHz	43 kHz	170 kHz	500 kHz
Maximum Pulse Repetition Rate ⁴	480 Hz	4 kHz	4 kHz	4 kHz	4 kHz

D.C. Voltage Measurement (Thermopile)

Signal Range ^{2,3}	0	1	2	3	4	5
Full-Scale Voltage	2.5mV	25.0 mV	250 mV	2.50 V	25.0 V	130 V
Resolution	76.3 nV	763 nV	7.63 μ V	76.3 μ V	763 μ V	3.96 mV
Accuracy (Filtered) ¹	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%
Accuracy (Maximum Measurement Rate=2 kHz)	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%
Bandwidth ⁶	3 kHz	3 kHz	3 kHz	3 kHz	3 kHz	3 kHz
Signal to Noise Ratio ¹	100dB	100dB	100dB	100dB	100dB	100dB

Pulse Voltage Measurement (Energy, Pyroelectric)

Signal Range ^{2,3}	0	1	2	3	4	5
Full-Scale Current	2.5mV	25.0 mV	250 mV	2.50 V	25.0 V	130 V
Resolution	76.3 nV	763 nV	7.63 μ V	76.3 μ V	763 μ V	3.96 mV
Accuracy	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%
Bandwidth ⁶	600 kHz	1000 kHz	1000 kHz	1000 kHz	400 kHz	1000 kHz
Maximum Pulse Repetition Rate ⁵	2 kHz	2 kHz	2 kHz	2 kHz	2 kHz	2 kHz

RMS Measurement

Maximum Pulse Repetition Rate	2 kHz	4 kHz
Accuracy	1.0%	2.0 %

Analog Output

Output Range	0	1	2
Full Scale Voltage (Load > 100 k Ω)	1 V	2 V	5 V
Accuracy	1.0 %	1.0 %	1.0 %
Linearity	0.3%	0.3%	0.3%

Trigger Level

Programmable Level	0...100 % Full Scale
Resolution	0.39 % Full Scale

¹ With 0.5 HZ filter on.

² Listed signal ranges specify meter capability. Available signal ranges are detector dependent.

³ Maximum measurable signal is detector dependent.

⁴ While the maximum repetition range may equal the bandwidth, it really depends on the signal duty-cycle or the signal shape.

⁵ The Maximum Repetition Rate refers to the meter pulse-by-pulse measuring capability. Due to its high bandwidth, the 1918-R can take in signals with higher repetitive rates and outputs them undistorted at the Analog Output.

⁶ The instrument bandwidth is determined by the detector used. Please refer to Newport Corporation's complete offering on detector type. The specified bandwidth is measured from the instrument input (detector) to the Analog Output BNC, using a generator with 50 Ω output.

2.4 Unpacking and Handling

It is recommended that the 1918-R Optical Meter be unpacked in a lab environment or work site. Unpack the system carefully; small parts and cables are included with the instrument. Inspect the box carefully for loose parts before disposing of the packaging. You are urged to save the packaging material in case you need to ship your equipment in the future.

2.5 Inspection for Damage

The 1918-R Optical Meter is carefully packaged at the factory to minimize the possibility of damage during shipping. Inspect the box for external signs of damage or mishandling. Inspect the contents for damage. If there is visible damage to the instrument or accessories upon receipt, inform the shipping company and Newport Corporation immediately.

**WARNING**

Do not attempt to operate this equipment if there is evidence of shipping damage or you suspect the unit is damaged. Damaged equipment may present additional hazards to you. Contact Newport technical support for advice before attempting to plug in and operate damaged equipment.

2.6 Available Options and Accessories

Newport Corporation also supplies temperature controlled mounts, lenses, and other accessories. Please consult with your representative for additional information.

2.7 Parts List



Figure 11 1918-R Optical Meter hard case

The following is a list of parts included in the hardcase (Figure 11) with the 1918-R Optical Meter:

1. A CD-ROM containing the User's Manual, Software Drivers and Utilities. (A PDF version of the manual can also be downloaded from the Newport website at www.Newport.com).
2. Battery Pack (1918-BAT).

3. External Power Supply (1918-PS) and detachable power cord (1918-PSC).

If you are missing any of these items or have questions about the items you have received, please contact Newport Corporation.

2.8 Choosing and Preparing a Suitable Work Surface

The 1918-R Optical Meter may be placed on any reasonably firm table or bench during operation. The kickstand can be rotated to two angles to improve visibility of the LCD display, as needed.

Provide adequate distance between the 1918-R Optical Meter and adjacent walls for ventilation purposes. Approximately 2-inch spacing for all surfaces is adequate.

2.9 Electrical Requirements

Before attempting to power up the unit for the first time, the following precautions must be followed:

- If the power supply is equipped with a 3-prong plug, connect the instrument to properly earth-grounded 3-prong receptacle only.



WARNING

If the power supply is equipped with a 3-prong plug, connect the instrument to properly earth-grounded, 3-prong receptacles only. Failure to observe this precaution can result in severe injury.

- Have a qualified electrician verify the wall socket that will be used is properly polarized and properly grounded.

2.10 Battery and External Power Supply



CAUTION

The battery needs to be charged for at least 8 hours before using the optical meter without the external power supply.

The 1918-R Optical Meter is shipped from the factory with the battery removed from the unit.

Install the battery before plugging the external power supply and before turning on the unit.

To install the battery first remove the battery compartment cover from the back of the unit. Connect the 5-pin battery connector to the corresponding connector inside the battery compartment (Figure 12). The connector is keyed and can only be inserted in one direction so please be careful to properly align the connector pins. Carefully fold the battery wires inside the battery

compartment so that the wires will not put pressure on the battery compartment cover. Close the cover and make sure it latches in place.

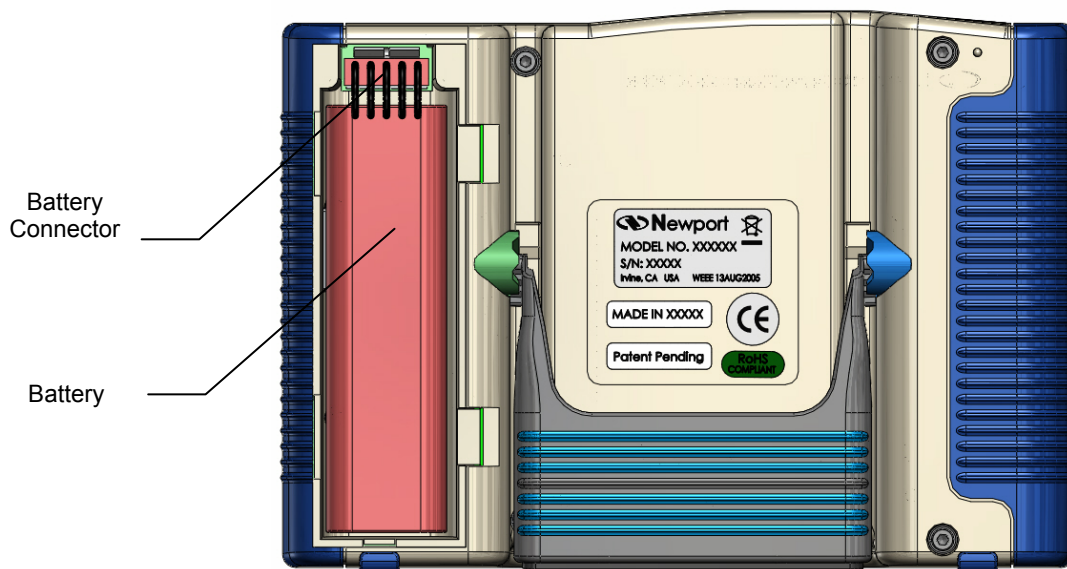


Figure 12 Rear panel with the battery cover removed

When the optical meter is turned on for the first time the battery needs to be charged for at least 4 hours. The optical meter can be used while the battery is being charged. It is recommended that the external power supply not be removed during the first 4 hours of charging.

	CAUTION
	The battery needs to be removed if the instrument is going to be unused for more than 3 weeks. Failure to do so may result in over-discharging the battery which drastically reduces the battery life or may cause battery breakdown.

AC power is supplied through an external power supply that provides in-line transient protection and RF filtering. This power supply is universal which means it can work at 90-264 VAC and 50/60 Hz.

	CAUTION
	Permanent damage may occur to the optical meter if external powers supply other than the Newport 1918-PS is used. Please call Newport Corporation if extra power supplies are needed for a particular setup.
	CAUTION
	Do not operate with a line voltage that is not within 90-264 VAC.

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3 System Overview

3.1 Startup Procedure

Provided that the Optical Meter has been installed in an appropriate environment and its external power supply power cord is connected to a working electrical outlet, power-up the Optical Meter by pressing the power button on the upper right corner of the left side panel. The battery will start charging. If the Optical Meter is turned on for the first time, make sure the external power supply is not unplugged from the AC outlet for at least 4 hours. During charging, the Optical Meter can be used for measurements. At the end of the charging time, the Optical Meter can be used without the external power supply.

For the highest precision and accuracy, the 1918-R Optical Meter should be allowed to warm up for one hour before being used for measurements.

3.2 Front Panel Layout



Figure 13 Front Panel Layout

3.2.1 Front Panel Elements

The front panel of the 1918-R features the following elements:

- A faceplate with a 4” active color liquid crystal display
- Setup/Enter and Esc keys
- Rubberized horizontal (left/right) and vertical (up/down) arrow keys
- Four rubberized blank keys below the display (referred to as “soft keys” later in the manual, as their function depends on the text on the screen above the key.)
- Six rubberized buttons with dedicated functions – Range, Mode, Hold, Filter, Lambda (λ), Zero.

3.2.2 Understanding the Main Screen

The main screen is displayed after startup (Figure 13). The middle portion of the main screen is a real-time display of power measured in last used units, and the lower half of the main screen is a row of four labels for actions that the keys below them will activate. Note that the screen will display “Saturated” message either when the amount of power surpasses the detector saturation specification, or when the detector spot size and/or detector saturation specifications are missing in the calibration module.

When an annunciator label is visible, its function is enabled. If the annunciator appears on the display as an unlabeled key, the function it represents is currently disabled. Annunciators loosely correspond to keypad keys, which are used either alone or in combination with the navigation and selection keys to control annunciator functions.

3.3 Side Panel Layout

Input Connectors

The input connectors are on the side panel. The 1918-R Optical Meter supports input from external detectors through a DB15 detector connector.

The external power supply is plugged in the DC INPUT connector.

Output Connectors

The 1918-R Optical Meter has an analog output, which enables direct monitoring of a detector through an oscilloscope or voltmeter.

3.3.1 Side Panel Layout

The side panel is a brushed aluminum plate with input and output connectors, a power switch, and LED indicators (Figure 14)

- Stand-by switch. This is a push-on/push-off button for turning the unit on or off.
- Ground Pin. The user can connect the Optical Meter to an Earth Ground for sensitive measurements.
- 15-Pin D-Sub Optical Detector Input.
- 3.5 mm Jack Analog Output.
- Mini USB Connector labeled USB PC. This connector is used for sending remote commands to the Optical Meter from a PC. The

connection to a computer is done via a USB cable Type "A" Male Connector to Type "Mini-B" Male Connector.

- USB "A" connector labeled USB MEM. This connector is used for saving the data on a USB Memory and for firmware upgrades.

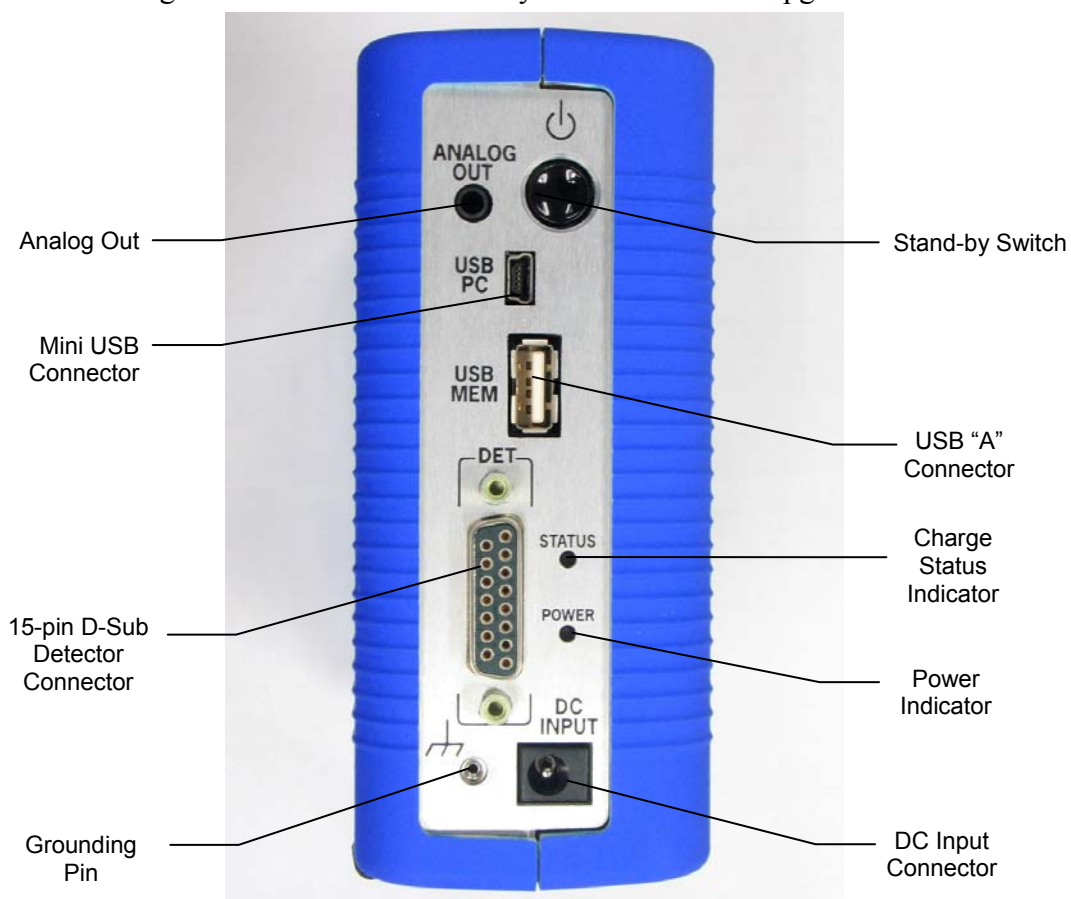


Figure 14 Side Panel Layout



WARNING

To avoid electrical shock hazard, connect the instrument to properly earth-grounded, 3-prong receptacles only. Failure to observe this precaution can result in severe injury.

3.4 Rear Panel Layout

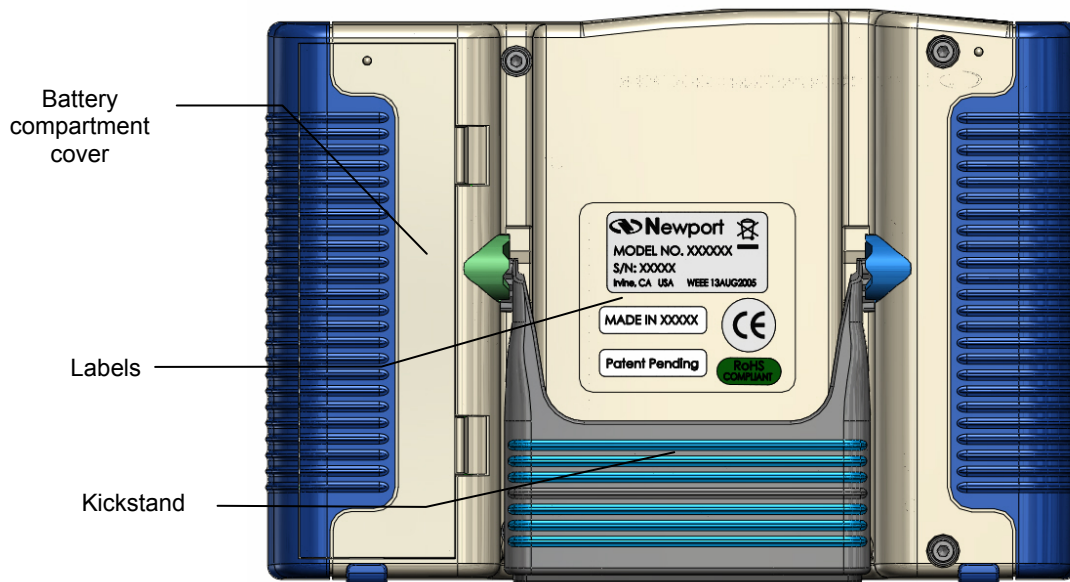


Figure 15 Rear panel

3.4.1 Rear Panel Elements

The rear panel of the 1918-R features the following elements (Figure 15):

- Battery compartment cover
- Kickstand
- Labels

Markings on the rear panel identify the instrument compliance with different standards and regulations.



CAUTION

There are no user-serviceable parts inside the Optical Meter. Work performed inside the Optical Meter by persons not authorized by Newport may void the warranty.

3.4.2 Kick Stand Positions

The instrument can be used in 4 viewing angles, allowing the user to adjust for the height of the work surface and lighting conditions:

- Laying on its back side in the horizontal position
- Standing up on its bottom side in the vertical position
- At two angled positions, by moving the kick-stand into two, preconfigured notches, manufactured into the instrument body.



Figure 16 Kickstand in the first angled position

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4 System Operation

4.1 Side Panel Power Switch and Indicators

4.1.1 Stand-by Switch



Figure 17 Power Switch

The Stand-by switch is located on the side panel and is a toggle button. It turns the unit on or in stand-by. When in stand-by, the unit is not fully turned off. In stand-by, the 1918-R Optical Meter will charge the battery, while all the other functions are off.

4.1.2 LED Indicators

The instrument has two LED indicators – one for Power and one for Status. Both of these are located on the Side Panel.

4.1.2.1 Power Indicator

The Power Indicator shows the current state of the battery charging circuit when the external power supply is powered and connected to the instrument. This indicator operates even when the unit is turned OFF. This indicator is not lit when the external power supply is disconnected from the instrument or is not connected to an external AC power source.

NOTE

The power indicator is not lit when the unit is operating from its internal battery.

When external power is applied, there are four states for this indicator – *Unlit*, *Flashing Red*, *Continuous Red*, and *Continuous Green*.

The Power Indicator will be *Continuously Green* when the battery charge circuitry detects a fully charged battery.

The Power Indicator will be *Continuously Red* while the battery is being charged.

The Power Indicator will be *Flashing Red* after external power is applied and until the battery charge circuitry determines that the instrument's internal battery is safe to charge. This may range from less than a second to several minutes, depending on the initial state of the battery.

Even when external power is applied to the instrument, the Power Indicator will not be lit when the charging circuitry detects a failed, defective, or missing battery. It is acceptable to operate the unit without a battery using just the external supply.

4.1.2.2 Status Indicator

The Status Indicator will switch every few seconds between *Green* and *Unlit* when the instrument is turned ON and is operating normally.

4.2 Front Panel Keys

The front panel keys are organized in three groups (see Figure 18).

- Navigation Keys
- Reconfigurable Keys (also called Soft Keys)
- Escape (ESC) Key

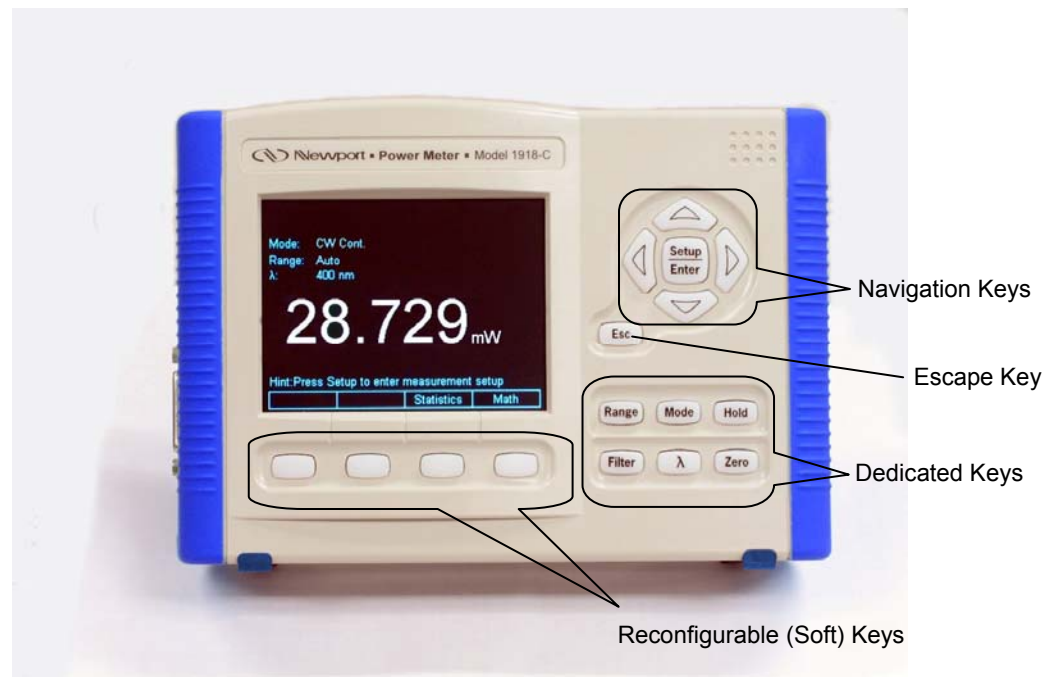


Figure 18 Front Panel keys

4.2.1 Setup/Enter Key

The Setup/Enter key is placed to the right of the display (Figure 19). This key has dual functionality. From the main screen (Figure 20) press this key to display the Measurement Settings screen (Figure 21). Also, use this key while

any secondary menu is displayed to complete entering data, or to exit the current secondary screen.



Figure 19 Navigation/Selection and Setup/Enter keys



Figure 20 Main screen



Figure 21 Measurement Settings screen

Refer to Section 4.3 for more information about the measurement settings.

4.2.2 Esc Key

The Esc key (Figure 18 and Figure 19) is used to cancel the current action. When in a secondary screen or menu, it will close the current screen or menu and the instrument will return to the main screen (Figure 20).

4.2.3 Navigation and Selection Keys

Navigation through and selection of data in the display is done with the top right group of four arrows keys and with the Setup/Enter key (Figure 19).

If the instrument is in Setup mode or in any configuration screens, pressing the arrow keys will select different setup modes as displayed by the current screen.

4.2.4 Soft Keys

Below the screen are a group of four keys (Figure 22). Their function varies, depending on the measurement mode or the setup screen. They are designed to provide context sensitive functionality to the user. The label displayed above each key indicates their function.



Figure 22 Soft keys

4.2.5 Dedicated Keys

Six dedicated function keys are at the bottom right portion of the front panel (Figure 23). Each of these keys can be used for quick access to the given function.



Figure 23 Dedicated keys

Range

Pressing this key reconfigures the Soft keys at the bottom of the screen as in Figure 24.

From this screen the user has two options.

- One is to toggle Auto/Manual Range mode. This is accomplished by pressing the left-most Soft key.

If the meter current configuration is Manual mode the left-most Soft key is labeled Auto to allow the user to change the Range selection in Auto Range Mode. This can be seen in Figure 24, where the range is 1.000 W indicating a manual mode.

If the instrument is in Auto Range Mode, the left most key is labeled Manual. The user may return to the main screen by pressing the ESC key (see Figure 25).

- The other option is to press the right-most Soft key, which is labeled Config. This action displays the Range Configuration screen. Here the user can select a certain range or even Auto Range with the Navigation/Enter keys. The number of ranges depends on the detector used (see Section 2.3.2). Once a range is selected the instrument returns to the main screen. While in Range Configuration screen, the user may cancel the selection by hitting the ESC key.



Figure 24 Manual Range mode



Figure 25 Auto Range mode



Figure 26 This screen is displayed after pressing the Config soft key

Mode

The Mode key displays a screen as in Figure 27. Using Navigation/Enter keys, the user can select different measurement modes or display modes according to his/her application. The ESC key cancels the selection and brings the instrument back to the main screen.

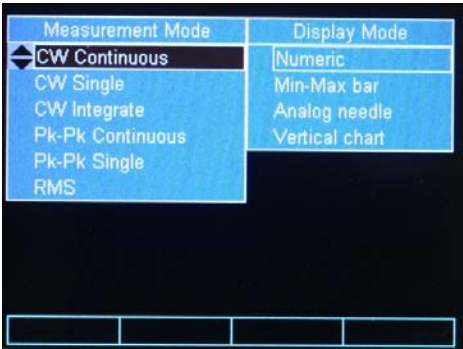


Figure 27 Mode selection screen

Hold

The Hold key toggles between holding the current measurement or running freely. When in Hold mode the numeric display freezes and the upper left corner of the display reads Hold (see Figure 28).

The display starts running when the Hold key is pressed again. The ESC key has no effect on Hold status.



Figure 28 Measurement is on hold

Filter

The Filter key allows the user to apply Analog or Digital Filters or both to the detector signal. This key brings a screen as in Figure 29. The Soft keys are reconfigured for the filter selection. Counting from left to right, the first key applies only the Analog filter to the detector signal, the second applies only the Digital Filter, and the third applies both filters. When the filters are selected, the corresponding label above the soft key has a highlighted background and the filter name is displayed in the upper left corner.



Figure 29 Filter selection screen

The fourth Soft key is used for filter configuration. When selected, a screen as in Figure 30 is displayed. Using the Navigation/Enter keys the user can select the filter of choice. If the Enter key is not pressed, the ESC key cancels the selection and brings the instrument back to the main screen. If the Enter key is pressed, a selection is made, and the instrument goes back to the main screen.

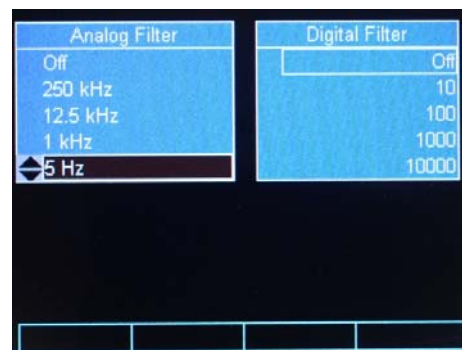


Figure 30 Filter configuration screen

The digital filter values can be edited (Figure 31). When one of the digital filter values is selected, the right most soft key becomes Edit Value. Pressing this Soft key displays a cursor on top of the first digit of the edited filter value. The Navigation Up/Down keys modify the digit, while the Left/Right keys move the cursor to the next digit. When finished, press the Enter key to store the new digital filter value and press Enter again to select the new value and exit the Filter configuration screen.



Figure 31 Digital Filter editing

Lambda (λ) Key

The Lambda (λ) key brings the Default and Custom Wavelengths screen (Figure 32). This screen allows the user to choose a default wavelength for the measurement in progress, or to choose a custom wavelength. See Section 4.6 for more details.

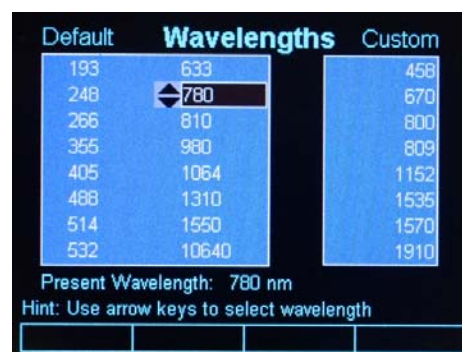


Figure 32 Wavelengths screen

Set Zero Offset (Zero Key)

The Zero Offset key, or short, Zero key is used to temporarily zero the instrument for the measurement in progress. When the user presses this key, the instrument takes the displayed numeric value as offset and subtracts it from all the subsequent measurements. When the instrument is zeroed the offset value is displayed on the main screen above the numeric value (see Figure 33).

The Zero key toggles the offset on or off. The ESC key has no effect on the Zero function.



Figure 33 Zero Selection

4.3 Measurement Settings

The Measurement Settings screen has dual functionality:

- One is the convenience the user has to change all the measurement settings within one screen.
- The other is information presented to the user about the detector used.

From the default screen, when pressing the Setup/Enter key, the 1918-R displays the Measurement Settings screen as in Figure 34.



Figure 34 Measurement Settings screen

4.3.1 Changing the Measurement Settings

Within the Measurement Settings screen the user can select the Wavelength, Range, Range Type, Beam Attenuator, Analog Filter selection, Digital Filter selection, Measurement Units, Measurement Mode, Spot Size and the Number of Digits displayed.

4.3.1.1 Wavelength Selection

With the Navigation/Selection keys bring the cursor on top of the Wavelength field. Hit the Enter key. A drop-down menu appears with the custom wavelength values (see Section 4.6 for more information about the wavelength custom values). Select the desired wavelength and hit Enter.

4.3.1.2 Range Selection and Range Type (Rng. type)

The Range field has dual functionality:

- One is to display the current selected range whether it was selected by the user, in Manual Range Mode, or by the system, in Auto Range Mode.
- The other is to allow the user to change the range.

With the Navigation/Selection keys bring the cursor on top of the Range field. Hit the Enter key. A drop-down menu appears with the available ranges in the selected units. Select the desired range and hit Enter.

If the unit was in Auto Range mode, once a range is selected here, it will switch the system to Manual Range Mode. To bring it back to Auto Range Mode, navigate to Range Type (Rng. type) and change the range to Auto Range Mode. Also, the user has another option to bring the Optical Meter back into the Auto Range Mode. This can be done from the default screen where the user can change the Range to Auto Range Mode with the Range Dedicated key (see Section 4.2.5).

4.3.1.3 Attenuator On/Off

If the detector is equipped with an integrated attenuator (e.g. 918D Series), the 1918-R detects its status (ON or OFF) and displays it in the Attenuator field.

The user has the option to manually set the attenuator to ON or OFF for detectors that have attenuators that can be manually mounted on the detector (e.g. 818 Series).

To change the attenuator status, bring the cursor on top of the Attenuator field. Hit the Enter key. A drop-down menu appears with the ON/OFF option. Select the setting and hit Enter.

4.3.1.4 Analog and Digital Filter Selection

With the Navigation/Selection keys bring the cursor on top of the Analog or Digital field. Hit the Enter key. A drop-down menu appears with the available filter settings. Select the desired filter and hit Enter.

The Analog and Digital filter configuration can be accessed any time with the Filter dedicated key. Refer to Section 4.2.5 for details.

4.3.1.5 Units Selection

With the Navigation/Selection keys bring the cursor on top of the Units field. Hit the Enter key. A drop-down menu appears with the available units. Select the desired units and hit Enter.

4.3.1.6 Mode Selection

This setting allows the user to change the measurement mode. The available modes are as follows:

- Continuous Wave Continuous Run (CW Cont.)
- Continuous Wave Single Shot (CW Single)
- Continuous Wave Integral (CW Integ.)
- Peak-to-Peak Continuous Run (Pk-Pk Cont.)
- Peak-to-Peak Single Shot (Pk-Pk Single)
- Pulse Mode Continuous Run (Pulse Cont.)

- Pulse Mode Single Shot (Pulse Single)
- RMS Measurements (RMS)

Refer to Section 5, Performing Basic Measurements, for detailed information regarding these modes.

4.3.1.7 Spot Size

This setting allows users to change the spot size. The default spot size is set to be same as the detector size. The spot size can be modified by navigating the cursor to this setting, and pressing the Setup/Enter key. The value then becomes editable; each digit in the value can be changed by pressing Up/Down arrow keys, and different digits can be selected by pressing Left/Right arrow keys. The desired value can be accepted by pressing Setup/Enter key.

4.3.1.8 Number of Digits Setting

The 1918-R displays a default 4 number of digits. To change these settings bring the cursor on top of the Num. Digits field. Hit the Enter key. A drop-down menu appears with the option of 3, 4 or 5 digits. Select the desired number of digits and hit Enter.

4.3.1.9 Offset Field

On the right column of the Measurement Settings the unit displays the Offset stored when the user hits the Zero dedicated key. This value can be cleared or updated if the unit is in the default screen and Zero is pressed (see Section 4.2.5 for more information).

4.3.2 Detector Information

The Measurement Settings screen displays information about the detector used based on the data available in the detector calibration module or detector internal memory.

The available data are the detector model number (Detector), the detector serial number (S/N), the detector temperature (Det. temp.) if the detector has an internal temperature sensor, and the detector responsivity (Responsivity).

4.4 Power Management

The Power Management Settings screen is used to select the instrument's power savings behavior.

Reducing the intensity of the LCD Display backlight significantly reduces the instrument's power consumption and thereby increases the time the instrument can be operated while powered from its internal battery. Power consumption is further reduced when the instrument is in its "Standby" mode.

The Power Management Settings screen can be accessed from the Measurement Settings screen (Figure 35) by pressing the Power soft key.

The Power Management screen is shown in Figure 36. The left column settings are applicable while the instrument is operating on battery power. The right column settings are applicable when the instrument is powered via the external power supply.



Figure 35 Accessing the Power Management screen

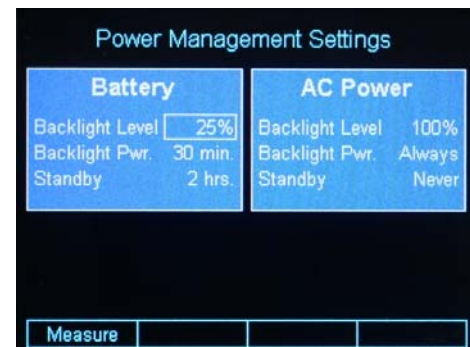


Figure 36 Power Management Settings screen

By default, the LCD Display Backlight is set for full (maximum) intensity. After a selected period of time without input from the keypad or communications via the USB interfaces, the instrument will enter its "Reduced Backlight" mode. In this mode the instrument functions normally; i.e., it is still capable of taking measurements; but the backlight is dimmer. The display will return to its normally bright intensity when any key on its keypad is actuated or when a command is received via the USB interface.

The Backlight Level can be changed from OFF to 100% in increments of 25%.

The Backlight Pwr. settings are used to select the time from the last keypad actuation, or USB command, until the instrument enters "Reduced Backlight" mode. The available settings are 1 minute, 5 minutes, 30 minutes, 2 hours, and Always. When "Always" is selected, the display is always at the maximum backlight intensity.

In Standby mode, measurements are disabled and the LCD Display Backlight is completely turned OFF, but the instrument will continue to monitor for keypad actuations and USB commands. When either of these "wake-up" events occurs, the instrument will read the detector; if the detector was not

changed while the instrument was in Standby mode, it will then revert to the user settings in effect when it entered Standby mode.

The Standby settings on the Power Management Settings screen are used to select the time from the last keypad actuation, or USB command, until the instrument enters Standby mode. The available settings are 1 minute, 5 minutes, 30 minutes, 2 hours, and never. When Never is selected, the instrument never enters Standby mode.

4.5 Trigger Setup

The Trigger Setup screen can be accessed from the Measurement Settings (Figure 37). Pressing the Trigger soft key gives the user a few choices to setup the trigger according to the measurement needs.

The 1918-R Series has an advanced Trigger set that allows the user to synchronize the measurements. Synchronization can be achieved with Trigger Start and Trigger Stop which can be set both from the trigger setup screen or with external commands.



Figure 37 Accessing the Trigger Setup screen

4.5.1 Trigger Start

The Trigger Start is an event that tells the system when to take a measurement or a group of measurements. It can be set in different ways.

- **Continuous.** The system is always triggered
- **Soft Key.** The system can be triggered with a Soft Key that appears in the main screen.
- **Command.** The system is triggered with an external command, PM:TRIG:STATE (Section 8.3.3).

The Trigger Start can also be set with an external command PM:TRIG:START (Section 8.3.3).

4.5.2 Trigger Stop

The Trigger Stop is an event that tells the system when to stop taking measurements. In the case of a single measurement, a trigger stop event arms the system for the next measurement. It can be set in different ways.

- **Never Stop.** The system measures continuously.
- **Soft Key.** The system stops measuring when a Soft Key is pressed. This key appears on the main screen.
- **Command.** The system stops measuring when an external command, PM:TRIG:STOP (Section 8.3.3) is sent via USB.
- **Value.** The system stops measuring when the measured signal crosses a user programmed value.
- **Time.** The system stops measuring when a user programmed time passed between the Start event and the current measurement.
- **Number of Samples.** The system stops measuring when a user programmed number of measurements passed between the Start event and the current measurement.

The Trigger Stop can also be set with an external command PM:TRIG:STOP (Section 8.3.3).

4.6 Wavelength Setting

Newport detectors have a calibration module or internal memory which stores the Responsivity versus Wavelength Table. If the Lambda key is pressed, a wavelength screen is displayed with the most common values in nanometers (nm) (see Figure 32 on page 41).

The screen has two columns. The left column displays the common wavelength values used in the industry, while the right column has custom values.

When the user selects one of the predefined wavelengths in the left column, the 1918-R looks up the wavelength in the responsivity table in the detector calibration module. If that exact value is found, the system will use the corresponding responsivity for that particular wavelength. If the value is not found, the system will calculate the responsivity using interpolation.

The right column in the Wavelength screen gives the user the option to set custom wavelengths. To do so, with the Navigation keys bring the cursor on top of one of the numbers. The rightmost Soft key becomes Edit Value. Hit this key to edit the custom number. A white cursor appears on the first digit. Press the Up/Down navigation keys to change the number and the Left/Right keys to move to the next digit. When finished, hit Enter. The new custom wavelength will be stored. Press Esc to return to the main screen

4.7 Display Color

In a laboratory environment, and especially when one uses protective eyewear, it may be desirable to change the meter display color to

accommodate the eyewear color. The instrument has predefined color schemes that can be changed any time (Figure 38).

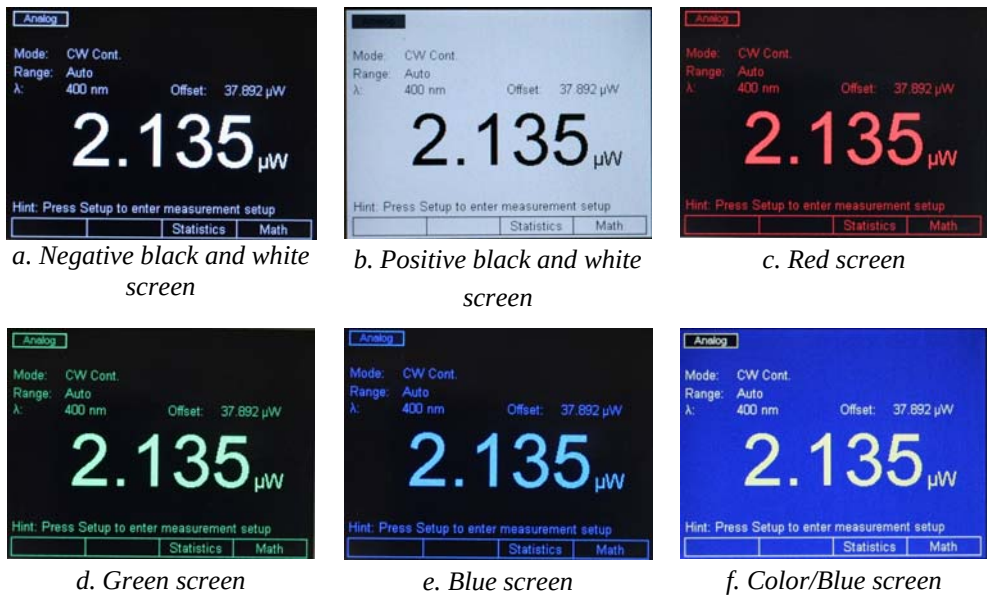


Figure 38 Predefined color schemes

To change the display colors first press the Enter/Setup key. This will bring the Setup screen and reconfigure the Soft keys (Figure 39).

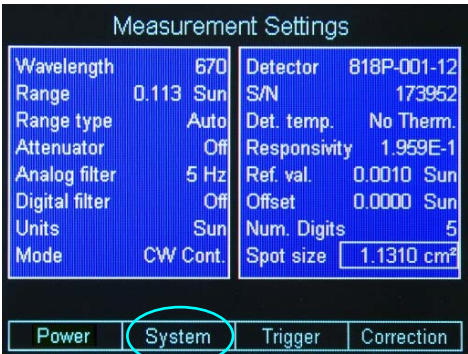


Figure 39 Setup screen

Pressing the System soft key will bring a secondary setup screen. Navigate to the Color Selection to change the screen color (Figure 40). Also, navigate to the Brightness field and hit Enter to change the screen brightness. The default value is 100%. Press the Measure soft key to go back to the Measurement Settings screen, or press ESC to go back to the main screen.



Figure 40 System Settings screen

Pressing the About soft key will bring a another screen with information about the unit Firmware version, serial number, calibration date. In addition, the attached detectors data is displayed (Figure 41). From here, one can navigate back to the Measurement Settings or back to the System Settings.

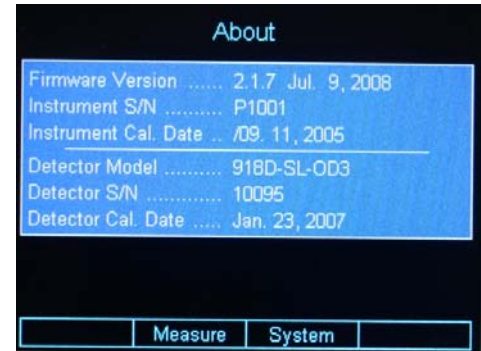


Figure 41 About screen

4.8 USB Address

The USB Address can be changed in the System Settings screen (Figure 40). Navigate to the USB Address and hit Enter. In the drop-down menu select the desired USB address.

4.9 Statistics

The 1918-R can display statistics for the measurement in progress. From the main screen press the Soft key labeled Statistics. The Statistics screen is displayed as in Figure 42.

The left column shows the statistics current setup. The Statistics function has two modes: Fixed and Continuous.

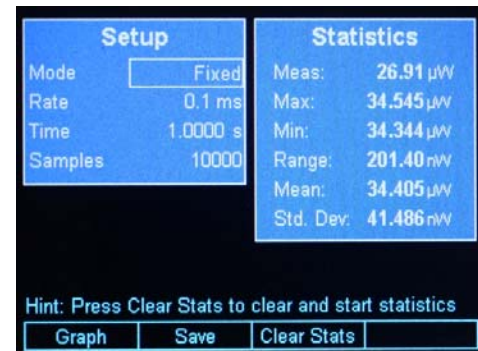


Figure 42 Statistics screen

When in Fixed mode the Statistics engine calculates the minimum (Min), maximum (Max), Range, Mean, Standard Deviation (Std. Dev.) for a fixed number of samples and displays them in the right column.

The number of samples is displayed in the left column on the Samples line. As Figure 42 shows, the number of samples is 10000 with a time interval between samples of 100 μ s. This means that the statistics values are calculated over 1 second interval.

If the Mode is Fixed, at the end of the 10000 samples, the statistics values are cleared and the calculation starts again with the next measurement.

The user can change the Mode with the Navigation keys. When the Enter key is pressed a drop-down menu allows the user to select Fixed or Continuous Mode. By using the Up and Down navigation keys followed by Enter, the user can change the Mode to Continuous.

Pressing Clear Stats soft key in Continuous mode clears current statistics,

enables data collection and starts computation of new statistics values. In this mode, the unit does not disable data collection when the number of samples specified is collected. Instead, it continues to collect data and place them in a data buffer on a first-in-first-out (FIFO) basis. The statistics shown are representative of data collected since the time Clear Stats was last pressed. As in Fixed mode, the Clear Stats soft key can be pressed at any time to restart the whole process again.

The Soft keys offer the user the option to clear the statistics at any time, with Clear Stats soft key, and to return back to the default screen with Display OFF soft key.

The Graph soft key (Section 4.9.1) can be used for the visualization of the measured values on a time graph.

The Statistics screen displays also the current measured value at the top of the right column.

4.9.1 Graph

The Graph soft key displays the last 10000 measurements on a time graph (Figure 43). Due to the sample time interval of 100 μ s, the graph duration is 1 second. The graph width is 270 pixels and, because of that, the system has to decimate the 10000 samples to fit them in this fixed number of pixels. Therefore, the graph might look choppy if the graph zoom is set on 1.



Figure 43 Graph screen

The user can zoom into the graph with the Soft keys +Zoom and –Zoom. As the zoom changes the graph displays a finer picture of the measurements. The zoom value is retained at the top of the screen. Pressing the +Zoom soft key will zoom in the graph 2 times the previous zoom value. Pressing the –Zoom soft key will zoom out ½ times the previous zoom value.

The maximum graph value is displayed in the upper left corner of the display. If the Cursor soft key is pressed, a vertical line (cursor) is displayed. The user can direct the cursor left or right with the Navigation Keys to read the measured values on the graph. The current value is displayed in the upper right corner and marked on the screen with a red dot.

4.10 Math Functions

The Math function is displayed on the math field, at the upper right corner, (Figure 44). To display the Math field the user selects the soft key labeled Math in the main screen. This in turn displays the math configuration screen

The Math function can be used to add, subtract, multiply or divide the current measurement Ch. A with a reference Ref. A in real time.

Once in the Math configuration screen the user can use the Navigation keys to build the mathematical expression. From the first column the user can select the first variable in the expression, from the second column, the operator, and from the third column, the second variable. When the selection was made, the system returns to the main screen after the Enter key was pressed.

The Ref. value is assigned a default value of 1 when the system is first turned on. The reference value is stored and displayed in the selected units. The user can change the reference value with a remote command

PM:REF:VAL *value*

as described in Section 8.3.

If the displayed units change, then the user needs to update the stored reference value so that the system will store the new units in the reference.

The Math field can be turned off from the math configurations screen. The second Soft key is labeled Display OFF. If this key is pressed the system returns to the default screen and the Math field disappears.



Figure 44 Math field

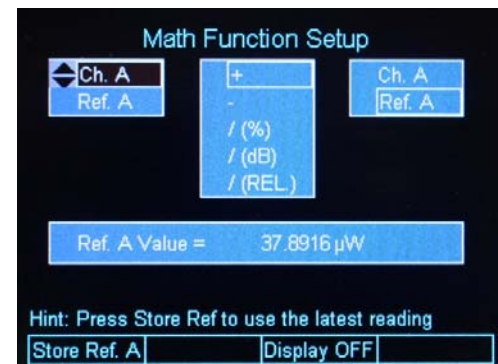


Figure 45 Math configuration screen

4.11 Measurement Correction Settings

The 1936/2936 Series power meters provide users the capability to correct actual measurements taken by it through a “Correction Settings” screen. This screen can be accessed by pressing the “Correction” soft-key in “Measurement Settings” screen (Figure 46). The “Measurement Settings” screen, as described earlier, can be accessed by pressing the Setup key from main measurement screen.

The Correction Settings screen (Figure 47) allows users to enter two (2) multiplier values, labeled “Multiplier 1” (M1) and “Multiplier 2” (M2), besides an “Offset” value. The corrected measurement is arrived at using the formula shown below:

Corrected measurement = [(Actual measurement * M1) – Offset] * M2.

The default value for the two multipliers is 1.0, and the offset is 0.0. If any one of these three parameters are changed from their default values, the main measurement screen shows the corrected measurement. A new field called “Detector:” appears at the bottom of the main screen that displays



Figure 46 Measurement Settings

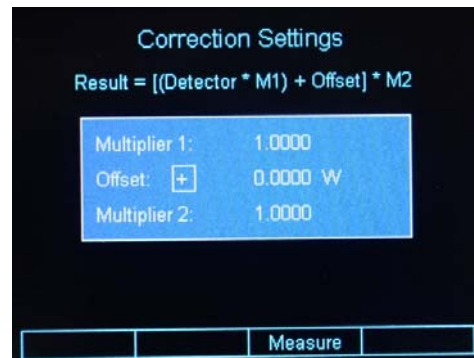


Figure 47 Correction Settings

4.12 Display Modes

The selection of various display modes can be done from the default screen, by pressing the Mode soft key. When the Mode selection screen is displayed (Figure 48), the display mode can be selected from the second column.

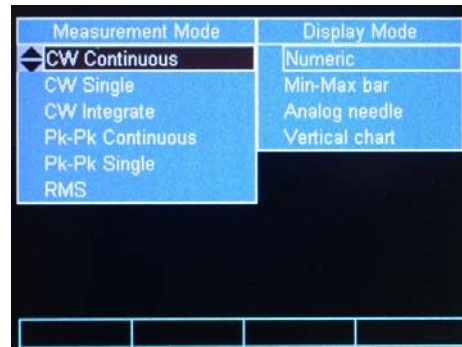


Figure 48 Mode selection screen

4.12.1 Numeric Display

The numeric display is the default display of the 1918-R. The numeric display shows the current measured value.



Figure 49 The Numeric display

4.12.2 Analog Bar

The Analog Bar display mode brings a bar graph below the numeric display. The bar graph is white and follows the value showed by the numeric display. The minimum and maximum labels displayed just below the bar graph represent the input range the unit is currently set on. For example, if the unit is set on 109.99 μW range, then the bar graph shows the same range of 0 to 109.99 μW . In the middle of the bar graph the displayed number is the middle of the range.



Figure 50 Analog Bar

The major ticks represent 10% of the range, and the minor ticks represent 5% of the range.

If the Soft key labeled Show Max is selected, the maximum value is retained and displayed in the bar graph with red color. The red bar is updated with each measurement, if the current measured value is larger than the largest of the previous measurements. Besides the maximum value being displayed with a red bar, the actual value is displayed above the bar graph where “Max=” shows the value. If the Soft key labeled Show Min is selected, the minimum value is retained and displayed in the bar graph with green color. The green bar is updated with each measurement, if the current measured value is smaller than the smallest of the previous measurements. Besides the minimum value being displayed with a green bar, the actual value is displayed above the bar graph where “Min=” shows the value.

The maximum and minimum bars can be reset with the Reset m/M soft key.

For users who would like to fine adjust a maximum or a minimum, the 1918-R offers the Auto Zoom function. When the Auto Zoom soft key is pressed another bar appears above the Analog Bar. The bar length is 2% of the Analog Bar. It is a display of the region around the end of the analog bar.

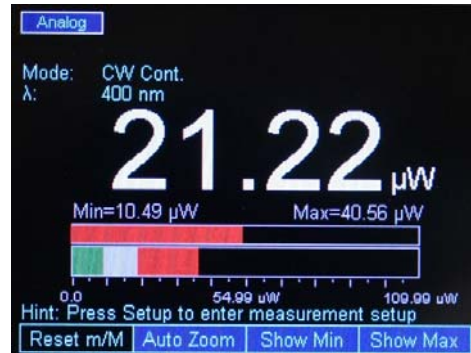


Figure 51 Auto Zoom

The Auto Zoom bar displays with white the current measured value, as the Analog Bar does. However its trip is more sensitive, because the maximum trip is $\pm 1\%$ of the Analog Bar. If the Show Max or Show Min soft keys are pressed, the system behaves the same as with the Analog Bar, displaying with red the maximum value and with green the minimum value. As an example, as Figure 51 shows, the red bar in the Analog Bar is very small, so with the Analog Zoom this bar is extended giving the user the convenience to see better small signal variations.

Pressing the Reset m/M soft key, clears the minimum and maximum bars from both the Analog Bar and the Auto zoom graphs.

4.12.3 Analog Needle

The Analog Needle displays a vertical marker that moves with the displayed numeric value. It is useful for users who look for a maximum or a minimum when adjusting the optical power.



Figure 52 Analog Needle display

4.12.4 Vertical Chart

When the Vertical Chart is selected the numeric display is moved in the upper right corner. The vertical chart scrolls down, representing a snapshot of the last measurements. The chart can be cleared with the Clear soft key. The maximum peak value of the chart is the maximum measurement of the entire chart, since the last clear. Below the chart, there is a line with the chart maximum and minimum in major units. On the same line in the middle, there is information about the chart zoom level, e.g. 2x, and the number of samples processed per chart line, e.g. 10 Avg (Figure 53).

The chart has 100 lines, each line having a white and red color. When the system fills up a line, it looks at a number of samples defined by the number displayed below the chart.



Figure 53 Vertical chart with 10 measurements per line



Figure 54 Vertical chart with 1 measurement per line

Figure 53 shows 100x -- 10 Avg, which means that, while painting a line, the system looks at the previous 10 measurements. It displays with white the minimum value in the 10- measurement list and with red the maximum value in the same list.

The user can change the number of measurements by selecting the Average soft key and then pressing the Right or Left Navigation keys, to increase, respectively decrease the number of measurements. If one measurement is selected, the chart color is white, because the maximum and minimum per line are the same (Figure 54). Due to one measurement display per line, the chart speed increases.

If the Zoom soft key is pressed the user can zoom into the chart with the Up and Down Navigation keys

4.13 Optical Meter Firmware Upgrade Procedure

Firmware Upgrade is an easy, straight-forward process. Simply copy firmware files (PM1918APP.EXE and XMLFILE1.XML) to a WinCE compatible USB Flash Drive and then plug it into the USB connector on the front of the instrument. Then wait a few seconds for the instrument to recognize the USB Flash Drive. The Optical Meter will detect the new firmware files and will ask if you want to download the files. Press the “Yes”

softkey to start the upgrade process. The Optical Meter will instruct you to restart once the upgrade is successful. Restart the Optical Meter by turning it OFF and back ON. The Optical Meter will restart running the new firmware.

New firmware files may be available either through the Newport web site (<http://www.newport.com>) at the product page or through your local Newport application specialist.

5 Performing Basic Measurements

5.1 Introduction

Many different types of optical measurements are possible using the 1918-R Optical Meter. Most of these possible measurements are selected from within the MODE menu. This chapter discusses these measurements.

5.2 Measurement Modes and Units

The 1918-R provides a number of measurement modes for acquiring data. At power on, the meter checks the detector's calibration EEPROM to determine the available measurement modes supported by the detector.

Measurements can be displayed in various units. The detector type and the measurement mode determine the set of units available at any given time. Table 1 illustrates the measurement modes and units available for each detector family. The measurement modes are grouped naturally into three categories: CW, Peak-to-Peak (Pk-Pk), and Pulsed. Each will be discussed in the sections below.

The 1918-R sets the measurement to a detector specific default mode depending on the detector used. All Newport detectors have internal logic or calibration modules. Based on the data stored in detectors, the instrument knows to auto configure itself and sets up the mode, range, filter, rate, etc. The user, in the setup screen, can change the detector specific default mode.

The basic measurement techniques for using the 1918-R are covered in the following sections. Refer to Table 1 for a review of the 1918-R's functions and measurement capabilities.

Detector Family	Mode	V	A	W	W/cm ²	J	J/cm ²	dBm	dB	REL / %	Sun
Low-Power (918D-Series)	CW Continuous		X	X	X			X	X	X	
	CW Single		X	X	X			X	X	X	
	CW Integrate					X	X		X	X	
	Pk-Pk Continuous		X	X	X			X	X	X	
	Pk-Pk Single		X	X	X			X	X	X	
High-Power (818P-Series)	CW Continuous	X		X	X			X	X	X	X
	CW Single	X		X	X			X	X	X	X
	CW Integrate					X	X		X	X	
Energy (818E-Series)	Pulse Continuous	X				X	X		X	X	
	Pulse Single	X				X	X		X	X	

Table 1 Available Measurement Modes and Valid Units.

The following instructions assume familiarity with the meter's functions. They also include steps to incorporate background correction and assume that the experimental setup under-fills and does not saturate or damage the detector.

5.3 CW Measurements (with 918D or 818P Detectors)

This section describes the procedure for making basic optical power measurements while properly removing the influence of ambient light and other drift effects.

- With a 918D (also 818-XX lower power detectors with proper adapters) or 818P-Series Detector connected to the meter, turn the meter on. Press the Mode key and then select CW Continuous with the Navigation keys. Use the ESC key to return to the Main screen. Set Range to Auto and press the Lambda (λ) key to set the measurement wavelength to the desired value.
- Cover or otherwise block the light source being measured and then press the Zero key to turn the Offset on. This effectively removes any background signal from subsequent measurements.
- Uncover the source so that it illuminates the detector and note the displayed value. This reading is the optical power observed by the detector due to the source.

This process assumes that the ambient signal is not changing between the time when the Zero key is pressed and when the measurement is made. The user should remember that, if he/she can see the detector active area as he/she moves around, then the detector registers this as a changing ambient DC signal!

- For the 918D series detectors (also 818-XX lower power detectors with proper adapters) you can decide whether to use the attenuator for your measurement or deactivate (or physically remove on 818 low power series) the attenuator.
 - a. For very low power measurements, below μW range in various ambient light environments, you can elect to use the detector with no physical attenuator in the optical path. This will increase the sensitivity and hence accuracy of the measurement.
 - b. In case of higher power measurement, the attenuator should be used to avoid damage or saturation of the detector. The attenuator use is recommended in mW to low Wattage range incident power. Please refer to the specifications of the particular detector to make sure you do not exceed the saturation levels of the detector.

For the 918D series detectors with integrated (non-removable) attenuators, there is a switch built into the detector head, sensing the position of the attenuator. The Optical Meter will then automatically use proper calibration data for presence or absence of the attenuator in front of the photo diode. For models such as 818 low power series detectors which have a removable attenuator, there is no switch built into the detector head, hence the user will have to manually select attenuator option status on the meter to obtain the proper power or signal readings.

5.4 Peak-to-Peak Power Measurements (918D Detectors)

This section describes the procedure for making basic optical peak-to-peak power measurements.

- With a 918D Low Power Detector connected to the meter, turn the meter on. Set the Mode to Pk-Pk Continuous. Set Range to Auto and press the Lambda (λ) key to set the measurement wavelength to the desired value.
- Illuminate the detector and note the displayed value. This reading is the peak-to-peak optical power observed by the detector.
- Accurate peak-to-peak power measurements can be made for pulse repetition rates up to 20 kHz.

5.5 Power Ratio and Power Reference measurements

This section describes how to use the mathematical functions in the Optical Meter to obtain a power or signal comparison to a previously saved value. The readout can be displayed as a linear ratio, a percentage or in dB. Refer to Section 4.10 for information on setting the reference values.

- **Example 1** – In order to select dB ratio versus a stored reference value:
 - a. While in the Math function screen, select channel A by using the up or down arrow keys. There will remain a black border around the highlighted/selected channel as you leave the column for the next selection.
 - b. Move to the next column by using the right arrow key, and select “/(dB)” option.
 - c. Move to the last column by using the right arrow key and use the up or down keys to select Ref. A.
 - d. Press the Menu/Enter key to accept the selections and exit the Math function screen.
 - e. The result of Ch. A/Ref. A in units of dB will appear in the designated window on the screen.
- **Example 2** – In order to get the difference between channel A and a stored reference value:
 - a. From the main screen of the meter, select soft key labeled “Math” to enter the math function screen. If you cannot see the label “Math”, press the “ESC” key.
 - b. Use the direction keys to highlight channel A in the first column
 - c. Using the direction keys, navigate to the “-“ symbol in the center column.
 - d. Using the direction key, move the final column such that Ref. A is highlighted.
 - e. Press the Menu/Enter key to retain the settings and exit the screen at the same time.
 - f. The result of Ch. A- Ref. A will be displayed in the designated window of the main display screen.

5.6 Pulse Energy Measurements (818E Detectors)

This section describes the procedure for making basic optical pulse energy measurements.

- With an 818E Energy Detector connected to the meter, turn the meter on. Set the Mode to Pulse Continuous. Set the Range to Manual Mode and press the Lambda (λ) key to set the measurement wavelength to the desired value.
- Illuminate the detector and note the displayed value as the meter measures each laser pulse. These readings represent the energies of the incident laser pulses. The meter will display the last pulse energy measured until a new pulse arrives.
- Accurate measurements can be made for pulse repetition rates up to 10 kHz, depending on the limitation of the specific detector in use, of course.

5.7 Signal Integration Measurements (918D or 818P Detectors)

This section describes the procedure for making a basic signal integration measurement while properly removing the influence of ambient light and other drift effects. The 1918-R begins and ends the signal integration every second.

- With a 918D or 818P Detector connected to the meter, turn the meter on. Set the Mode to CW Continuous. Set the Range to Auto and press the Lambda (λ) key to set the measurement wavelength to the desired value.
- Cover or otherwise block the light source being measured, turn Zero on and then set the Mode to CW Integrate. Immediately upon entering the integration mode, the meter will begin to acquire and integrate data. The display value may reflect the integration of noise due to ambient temperature fluctuations (when using the 818P detectors) or light fluctuations (when using the 918D detectors).
- Uncover or trigger the source. The displayed value should now reflect the detector signal integration value.

NOTE

This process assumes that ambient signals are not changing between the moments when the user zeros the display and when the measurement is made.

5.8 Measuring Laser Pulse Energy with an 818P Thermopile Detector (Single Shot)

This application makes use of the 1918-R' CW Integrate mode (see Section 5.7 above). When an optical pulse with energy $E(\lambda)$ is incident on a thermopile (818P Series), a voltage signal is generated at the detector input as the heat pulse flows out to the cooling fins. The sum of a series of integrated signals (each one second long) resulting from this heat pulse is a measure of the optical pulse energy, see Figure 55.

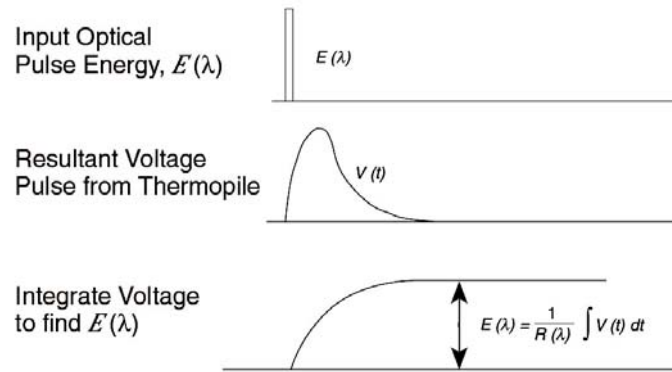


Figure 55 Measuring Laser Pulse Energy via a Thermopile in CW Integrate Mode

A recommended procedure is:

- With an 818P High Power Detector connected to the meter, turn the meter on. Set the Mode to CW Continuous. Set the Range to Auto and press the Lambda (λ) key to set the measurement wavelength to the desired value. Set Zero on and then change Mode to CW Integrate.
- Before the optical pulse arrives, the display may reflect the integral of detector noise due to ambient temperature fluctuations.
- Trigger the laser pulse. The display will display a new integrated energy measurement every second. The readings will start decreasing rapidly after the first couple of measurements due to the decline in heat flow from the relatively slow thermopile detector.
- When the displayed reading drops again to the point of essentially displaying the detector's noise component, the individual stored readings may be retrieved via the USB interface. Summing these readings will yield the pulse energy.

NOTE

This method works best if the integrated result of the pulse signal is much larger than the integral of the detector's noise component. If the integrated result of the pulse is not much larger, then error in the measurement will arise due to the uncertainty generated by integration of the noise component terms.

NOTE

The time constant of a thermopile detector determines the amount of time that one should expect to wait when making an integrated energy measurement of an optical pulse. Typically, an accurate value will be at 5 time constant after the arrival.

5.9 RMS Measurements

This section describes the procedure for making a basic signal RMS (Root Mean Square) measurement while properly removing the influence of ambient light and other drift effects. The 1918-R begins and ends the signal RMS every second.

- With a 918D or 818P Detector connected to the meter, turn the meter on. Set the Mode to RMS. Set the Range to Auto and press the Lambda (λ) key to set the measurement wavelength to the desired value.
- Cover or otherwise block the light source being measured, turn Zero on. Immediately upon entering the RMS mode, the meter will begin to acquire and calculate RMS value of the data. The display value may reflect the RMS of noise due to ambient temperature fluctuations (when using the 818P detectors) or light fluctuations (when using the 918D detectors).
- Uncover or trigger the source. The displayed value should now reflect the detector signal RMS value.
- The RMS value is calculated based on data acquired over a one second time period. The formula used to calculate the RMS value is given below:

$$y = \sqrt{\frac{\sum_{i=0}^n x_i^2}{n}}$$

Where x_i is the signal measured, n is the number of samples acquired over a one second period, and y is the RMS value.

NOTE

This process assumes that ambient signals are not changing between the moments when the user zeros the display and when the measurement is made.

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6 Software Application

6.1 Overview

The 1918-R has a USB connector on the side of the unit that is used to connect to a computer for use with this application. (RS-232 connection for this application is not supported.)

Provided on the CD that comes with the unit is an installation for this software application, it communicates with the 1918-R using the USB port. The installation installs the USB drivers that are required to use USB communication.

The design of the software is to allow the user to remotely control basic functions of the instrument.

6.2 Connection

Start the application, which will detect and connect the attached Optical Meter. The 1918-R Optical Meter has just one channel, so channel B is grayed out.

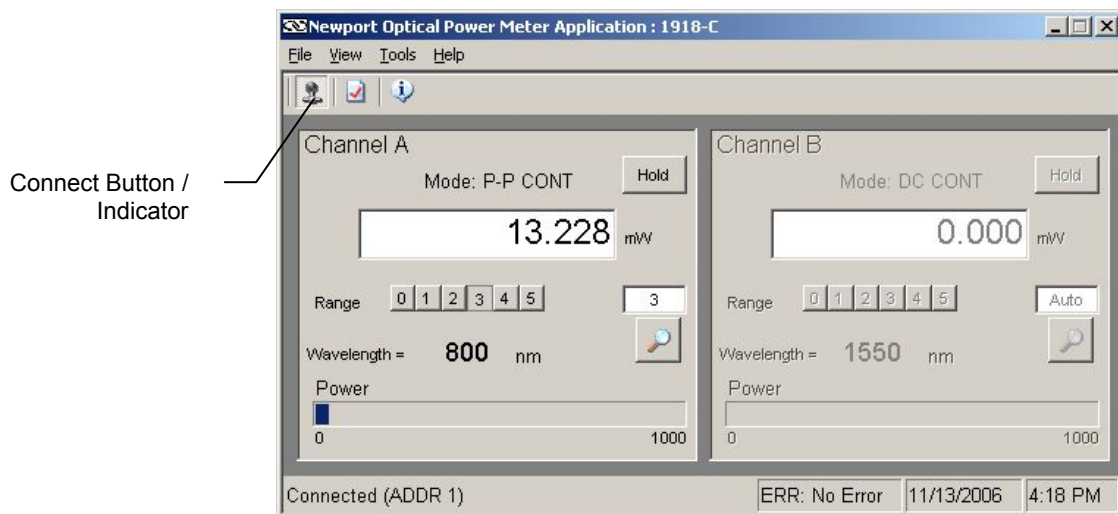


Figure 56 Application front panel

6.3 General Usage

This software application allows the user to setup and monitor the instrument remotely.

The controls on the instrument are available in the software in a very easy to read and change format.

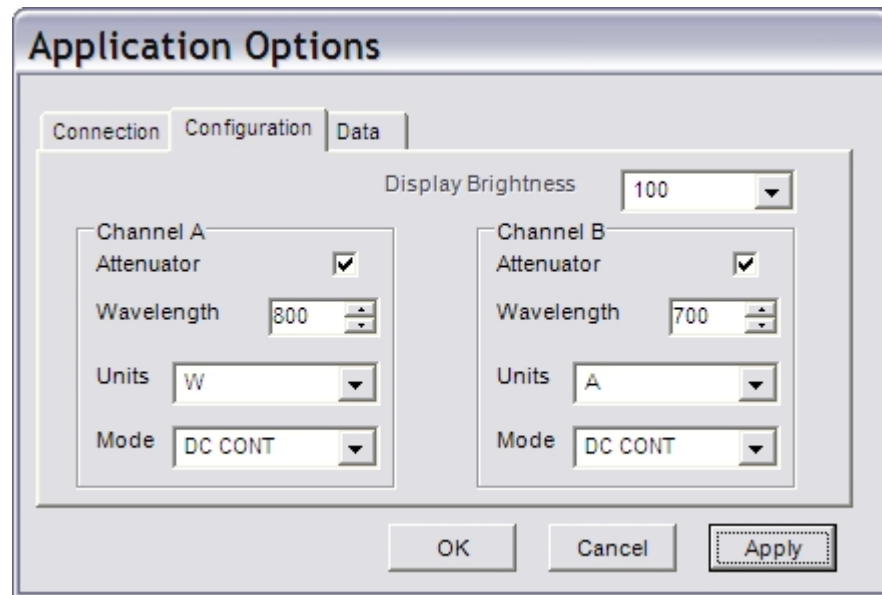


Figure 57 Application Advanced Options (Configuration Tab)

The application is designed to have menus similar to standard Windows applications like MS Word, to ease usability.

6.4 Menu Structure

To Exit the application go to the File menu and select Exit.

The Edit/Advanced Options menu has additional property settings, including channel settings and data logging options.

The Help/About will show information about the application, including firmware version when an instrument is connected and communicating.

7 Computer Interfacing

7.1 General Guidelines

The 1918-R Optical Meter has a USB computer interface port used to send commands to the Optical Meter from a host PC. The connection to a computer is done via a USB cable Type "A" Male Connector to Type "Mini-B" Male Connector.

The commands supported by the Optical Meter can be divided into the following two categories: commands that cause the Optical Meter to take a desired action, and commands (queries) that return a stored value or state of the Optical Meter.

Query commands are always terminated by a question mark (?). It is recommended that when a query command is sent, the response to that command from the Optical Meter be read before issuing any other command.

Set commands, on the other hand, are used to configure/setup the Optical Meter for a desired mode of operation. These commands take at least one parameter. The subsequent sections in this chapter detail the USB communication protocols supported by the Optical Meter.

7.2 Computer Interface Terminology

Listed below are the key abbreviations and concepts used in the command reference section (Section 8) of this manual.

7.2.1 <...> Delimiting Punctuation

For the purposes of this manual, any string enclosed by <...> is considered to be a command, a string or numerical argument. The punctuation <...> is used to symbolize the typographic limits of the command, string or argument in question.

7.2.2 <CR> Carriage Return

The ASCII encoded byte 13 in decimal. (0D hex)

7.2.3 <LF> Line Feed

The ASCII encoded byte 10 in decimal. (0A hex)

7.2.4 (;) Semicolons

Semicolons are used to separate commands within a single transmission (concatenation).

7.2.5 <number> Numerical Types

Numerical parameters are passed and returned as the actual ASCII characters in the string representation of the number. See section 12.2 for more detailed information.

7.2.6 <string> String Types

See the section 12.1 for a detailed description of <string>.

7.3 USB Communication

The instrument is designed to communicate with standard USB interfaces. Before connecting the instrument to the USB interface the user should install the application included in the software CD that accompanies the Optical Meter. The application automatically installs the right USB drivers. Communication can be done through this interface by using the application or by developing software in the user's preferred programming language. The software CD contains drivers and example programs in the following programming languages: LabVIEW, Visual Basic, and Visual C++.

8 Communication Command Reference

8.1 1918-R Optical Meter Remote Interface Commands

A complete listing of the commands supported by the 1918-R Optical Meter is provided below.

8.2 Command Overview

There are two types of commands: commands that cause the Optical Meter to take a desired action, and queries that return a stored value or state of the Optical Meter. Queries must end with a question mark (?), while commands may require parameter(s) to follow:

PM:Lambda 810

For example, the value “810” in the command **PM:Lambda 810** sets the wavelength to 810nm. The table below summarizes all the commands and queries supported by the 1918-R Optical Meter. The command/query **MUST** contain all of the letters, which are shown in upper case in this table. The lower case letters shown with the commands are optional, and may be used for clarity. If any of the optional letters are used, then all of the optional letters are required for the command.

The commands may be sent to the instrument in either upper or lower case or in any combination. For example, the following commands are equal:

PM:Lambda 810

PM:L 810

pm:L 810

Pm:L 810

COMMAND EXECUTION:

The controller interprets the commands in the order they are received and execute them sequentially. If a set of commands have to be executed closer to each other, these commands can be sent to the controller simultaneously by creating a command string with semicolon (;) used as a command separator. The command string length should not exceed 50 characters. In the example shown below, a command string was created with semicolon

separating 5 queries. The controller responds to this command string with a response that has 5 values using a comma (,) as a separator.

COMMAND STRING:

PM:P?;PM:ATT?;PM:L?;ERR?

INSTRUMENT RESPONSE:

1.2450,1,810,0

8.3 Optical Meter Command Description

8.3.1 Command Glossary

Root level Commands/Queries Summary

Name	Number of Parameters	Function	Page No.
*IDN?	NONE	Identification query	74
*RCL	1	Recall configuration settings	74
*SAV	1	Save configuration settings	74
ADDRes	1	Sets the instrument's USB address	75
ADDRes?	NONE	Returns the instrument's USB address	75
ERRors?	NONE	Returns errors generated since the last query.	76
ERRSTR?	NONE	Returns errors and their corresponding error text generated since the last query.	76

Table 2 Root level Commands/Queries Summary

Tree level Commands/Queries Summary

Name	Number of Parameters	Function	Page No.
DISP:BRIGHT	1	Sets the backlight level of the display and the keypad	72
DISP:BRIGHT?	NONE	Returns the backlight level of the display and the keypad	72
PM:AC:BackLight	1	Sets the inactive time duration for reducing the backlight level during AC operation	76
PM:AC:BackLight?	NONE	Returns the inactive time duration for reducing the backlight level during AC operation	77
PM:AC:LEVEL	1	Sets the reduced backlight level following inactive time duration during AC operation	77
PM:AC:LEVEL?	NONE	Returns the reduced backlight level following inactive time duration during AC operation	78
PM:AC:STANDBY	1	Sets the inactive time duration for going to standby during AC operation	78
PM:AC:STANDBY?	NONE	Returns the inactive time duration for going to standby during AC operation	78
PM:ANALOGFILTER	1	Sets the analog filter to desired value	79
PM: ANALOGFILTER?	NONE	Returns the analog filter setting	79
PM:ANALOG:IMP	1	Sets the analog output impedance to desired value	80
PM:ANALOG:IMP?	NONE	Returns the analog output impedance value	80
PM:ANALOG:OUT	1	Sets the analog output range to desired level	80
PM:ANALOG:OUT?	NONE	Returns the analog output range	81
PM:ATT	1	Selects if the attenuator's calibration data is included for power calculation.	81
PM:ATT?	NONE	Returns setting if attenuator data should or should not be used when calibrating the Optical Meter.	81
PM:ATTSN?	NONE	Gets the attenuator serial number.	82
PM:AUTO	1	Sets the Optical Meter ranging to manual or automatic.	82
PM:AUTO?	NONE	Returns 1 if automatic Optical Meter ranging is selected.	82

Name	Number of Parameters	Function	Page No.
PM:BAT:BL	1	Sets the inactive time duration for reducing the backlight level during battery operation	82
PM:BAT:BL?	NONE	Returns the inactive time duration for reducing the backlight level during battery operation	83
PM:BAT:LEVEL	1	Sets the reduced backlight level following inactive time duration during battery operation	83
PM:BAT:LEVEL?	NONE	Returns the reduced backlight level following inactive time duration during battery operation	84
PM:BAT:STANDBY	1	Sets the inactive time duration for going to standby during battery operation	84
PM:BAT:STANDBY?	NONE	Returns the inactive time duration for going to standby during battery operation	85
PM:CALDATE?	NONE	Returns the calibration date of the detector.	85
PM:CALTEMP?	NONE	Returns the temperature at which the calibration was performed.	85
PM:CORR	3	Sets the power measurement correction settings.	86
PM:CORR?	NONE	Power measurement correction settings query	86
PM:DETMODEL?	NONE	Returns the model number of the detector.	86
PM:DETSIZE?	NONE	Returns the detector surface area	87
PM:DETSN?	NONE	Returns the serial number of the detector.	87
PM:DIGITALFILTER	1	Sets the digital filter to desired value	87
PM: DIGITALFILTER?	NONE	Returns the digital filter setting	87
PM:DPower?	NONE	Detector Power query	88
PM:DS:BUfFer	1	Set data store behavior select.	88
PM:DS:BUfFer?	NONE	Returns data store behavior select.	88
PM:DS:Clear	NONE	Clear data store.	89
PM:DS:Count?	NONE	Returns data store count of items stored.	89
PM:DS:Enable	1	Set data store enable.	89
PM:DS:Enable?	NONE	Returns data store enable.	89
PM:DS:GET?	1	Returns data store data. {1,1-10,-5,+5} – value, range, oldest 5, newest 5	90
PM:DS:INTerval	1	Set data store interval.	90
PM:DS:INTerval?	NONE	Returns data store interval.	90
PM:DS:SAVEBUFER	1	Saves the data store buffer to a WinCE compatible USB flash disk	91
PM:DS:SIZE	1	Sets the size of the Data Store buffer	91
PM:DS:SIZE?	NONE	Returns the sizes of the Data Store buffer	91
PM:DS:UNITs?	NONE	Returns data store units.	92
PM:FILTer	1	Selects the filtering operation: no filtering, analog filter, digital filter, or analog and digital.	92
PM:FILTer?	NONE	Returns the filtering operation: no filtering, analog filter, digital filter, or analog and digital.	92
PM:Lambda	1	Sets the wavelength for use when calculating power.	93
PM:Lambda?	NONE	Gets the selected wavelength in nanometers.	93
PM:MAX:Lambda?	NONE	Returns the longest calibrated wavelength in nanometers.	93
PM:MAX:Power?	NONE	Returns the maximum readable power in present range	94
PM:MIN:Lambda?	NONE	Returns the shortest calibrated wavelength in nanometers.	94
PM:MIN:Power?	NONE	Returns the minimum readable power in present range	94
PM:MEAS:TIMEOUT	1	Sets the measurement timeout period	94
PM:MEAS:TIMEOUT?	NONE	Returns the measurement timeout period	95
PM:MODE	1	Acquisition mode select	95
PM:MODE?	NONE	Returns the currently selected acquisition mode.	96
PM:Power?	NONE	Returns the power in the selected units.	96
PM:PWS?	NONE	Returns the power with status.	96
PM:RANge	1	Selects the gain stage when making readings with the detector head within a range from 0 to 6 (with zero being the highest).	97
PM:RANge?	NONE	Returns an integer indicating the current range.	98
PM:REF:VALue	1	Sets the user reference value for use in relative or dB readings.	98

Name	Number of Parameters	Function	Page No.
PM:REF:VALue?	NONE	Returns the user reference value.	98
PM:REF:STOre	NONE	Sets the user reference value for use in relative or dB readings as the present reading.	98
PM:RESPOnsivity?	NONE	Gets the responsivity currently used for making power calculations.	99
PM:RUN	1	Disables or enables the acquisition of data.	99
PM:RUN?	NONE	Returns the present acquisition mode.	99
PM:SATLEVEL	1	Set saturation current density or power level	100
PM:SATLEVEL?	NONE	Query saturation current density or power level	100
PM:SPOTSIZE	1	Sets the detector spot size	100
PM:SPOTSIZE?	NONE	Returns the detector spot size	100
PM:STAT:MAX?	NONE	Returns statistics buffer maximum value.	101
PM:STAT:MEAN?	NONE	Returns statistics buffer mean value.	101
PM:STAT:MIN?	NONE	Returns statistics buffer minimum value.	101
PM:STAT:MAXMIN?	NONE	Returns statistics buffer maximum-minimum value.	101
PM:STAT:SDEVIation?	NONE	Returns statistics buffer standard deviation value.	102
PM:Temp?	NONE	Returns the 918 detector's temperature in degrees Celsius.	102
PM:THERM:PREDICT	1	Disables or enables thermopile prediction algorithm	102
PM:THERM:PREDICT?	NONE	Returns the present thermopile prediction algorithm state	103
PM:TRIG:START	1	Set the optional start event	103
PM:TRIG:START?	NONE	Returns optional start event	103
PM:TRIG:STOP	1	Set the optional stop event	103
PM:TRIG:STOP?	NONE	Returns optional start event	104
PM:TRIG:STATE	1	Set the trigger state	104
PM:TRIG:STATE?	NONE	Returns the trigger state	104
PM:UNITs	1	Selects the units for readings.	105
PM:UNITs?	NONE	Returns an integer indicating the selected units.	105
PM:ZEROSTOre	NONE	Sets the zeroing value with the present reading.	106
PM:ZEROVALue	1	Sets the zeroing value.	106
PM:ZEROVALue?	NONE	Gets the zeroing value.	106

Table 3 Tree Level Commands/Queries Summary

8.3.2 Display Commands

8.3.2.1 DISP:BRIGHT

Description	Display brightness command	
Syntax	BRIGHT <i>brightness</i>	
Remarks	The BRIGHT command controls the brightness of the instrument display.	
Argument	Type	Description
<i>Brightness</i>	int	Brightness, in levels from 0 to 7
Related Commands: DISP:BRIGHT?		

8.3.2.2 DISP:BRIGHT?

Description	Display brightness query		
Syntax	DISP:BRIGHT?		
Remarks	The BRIGHT? query returns the display brightness setting.		
Response	Type	Description	

Brightness int Display brightness, in levels from 0 to 7

Related Commands: **DISP:BRIGHT**

8.3.3 Optical Meter Commands

8.3.3.1 *IDN?

Description Identification Query

Syntax *IDN?

Parameters None

Function

Remarks This query will cause the Optical Meter to return an identification string.

Model name	Firmware version #	Firmware date	Controller Serial #
┌────────┐	┌────────┐	┌────────┐	┌────────┐
NEWPORT XXXX vYYY mm/dd/yy SNZZZZ			

Examples:

NEWPORT 1918-R v2.0.0 11/07/06 SN0001

8.3.3.2 *RCL

Description Recall Configuration Settings

Syntax *RCL *bin*

Remarks The *RCL command restores the power meter to the setup state saved in its non-volatile flash memory.

Argument	Value	Description
<i>bin</i>	0	Reserved
	1 to 5	Valid configuration settings

Related Commands: *SAV

8.3.3.3 *SAV

Description Save Configuration Settings

Syntax *SAV *bin*

Remarks The *SAV command saves the present state of the power meter in its non-volatile flash memory. A particular state is then recalled using the *RCL command. If any one of these parameters are changed, the present state of the power meter will automatically be saved in configuration setting #1. When the power meter is reset, the state of the meter defaults to configuration setting #1. The setup parameters saved include:

Display brightness level

USB address

Color scheme
 Measurement display mode
 Custom wavelengths
 AC backlight level
 AC backlight power
 AC standby
 Battery backlight level
 Battery backlight power
 Battery standby

Argument	Value	Description
<i>bin</i>	<i>0</i>	<i>Reserved</i>
	<i>1 to 5</i>	<i>Valid configuration settings</i>
Related Commands: *RCL		

8.3.3.4**ADDRess**

Description USB address command.

Syntax **ADDRess** value

Remarks **The ADDRess** command sets the Optical Meter USB address. After changing USB address, the communication with the Optical Meter has to be re-initialized.

Argument	Value	Description
<i>Value</i>	<i>0</i>	<i>Reserved</i>
	<i>1 to 31</i>	<i>Valid USB address range</i>

Related Commands: **ADDRess?**

8.3.3.5**ADDRess?**

Description USB address query.

Syntax **ADDRess?**

Remarks **The ADDRess** query returns the Optical Meter's USB address.

Response	Value	Description
<i>address</i>	<i>0</i>	<i>Reserved</i>
	<i>1 to 31</i>	<i>Valid USB address range</i>

Related Commands: **ADDRess**

8.3.3.6**ERRors?****Description**

Error query

Syntax**ERRors?****Remarks**

The **ERRors?** query returns a single error number that has occurred since the last query. This error is indicated by a number that corresponds to the type of error that occurred. This command also clears the read error from the Error buffer.

Response**Type****Description***Error code*

int

Error code number per Appendix B, 0 if no errors

Related Commands: **ERRSTR?****8.3.3.7****ERRSTR?****Description**

Error string query

Syntax**ERRSTR?****Remarks**

The **ERRSTR?** query returns a single error number along with the corresponding error text string that have occurred since the last error query.

Response**Type****Description***Error code,*

string

Error code number and text for error code as "*text*" per Appendix B, 0 if no errors**Related Commands:** **ERRors?****8.3.3.8****PM:AC:BackLight****Description**

Inactive time period for reducing the backlight level during AC operation

Syntax**PM:AC:BackLight** *value***Remarks**

If the instrument is on AC power, its display backlight will be reduced to a pre-specified level following a period of inactivity specified using this command. The instrument is considered to be inactive if either the keys on the front panel are not being pressed or if there is no communication with a host PC over USB interface.

Argument**Value****Inactive Time Period***Value*

0

1 min

1

5 min

2

30 min

3	2 hours
4	Always

Related Commands: **PM:AC:BackLight?**; **PM:AC:LEVEL?**

8.3.3.9 **PM:AC:BackLight?**

Description Inactive time period for reducing the backlight level during AC operation

Syntax **PM:AC:BackLight?**

Remarks The **PM:AC:BackLight?** query returns an integer indicating the present inactive time period for reducing the backlight level during AC operation.

Argument	Value	Inactive Time Period
<i>Value</i>	0	1 min
	1	5 min
	2	30 min
	3	2 hours
	4	Always

Related Commands: **PM:AC:BackLight**; **PM:AC:LEVEL**

8.3.3.10 **PM:AC:LEVEL**

Description Display backlight level following an inactive time period during AC operation

Syntax **PM:AC:LEVEL** *value*

Remarks If the instrument is on AC power, its display backlight will be reduced following a pre-specified period of inactivity to the level specified using this command. The instrument is considered to be inactive if either the keys on the front panel are not being pressed or if there is no communication with a host PC over USB interface.

Argument	Value	Backlight Level
<i>Value</i>	0	100%
	1	75%
	2	50%
	3	25%
	4	OFF

Related Commands: **PM:AC:BackLight?**; **PM:AC:LEVEL?**

8.3.3.11 PM:AC:LEVEL?

Description Display backlight level following an inactive time period during AC operation

Syntax **PM:AC:LEVEL?**

Remarks The **PM:AC:LEVEL?** query returns an integer indicating the display backlight level following an inactive time period during AC operation.

Argument	Value	Backlight Level
<i>Value</i>	0	100%
	1	75%
	2	50%
	3	25%
	4	OFF

Related Commands: **PM:AC:BackLight**; **PM:AC:LEVEL**

8.3.3.12 PM:AC:STANDBY

Description Inactive time period for putting the instrument in standby mode during AC operation

Syntax **PM:AC:STANDBY** *value*

Remarks If the instrument is on AC power, it will be put in standby mode following a period of inactivity specified using this command. The instrument is considered to be inactive if either the keys on the front panel are not being pressed or if there is no communication with a host PC over USB interface.

Argument	Value	Inactive Time Period
<i>Value</i>	0	1 min
	1	5 min
	2	30 min
	3	2 hours
	4	Never

Related Commands: **PM:AC:STANDBY?**; **PM:AC:LEVEL?**

8.3.3.13 PM:AC:STANDBY?

Description Inactive time period for putting the instrument in standby mode during AC operation

Syntax **PM:AC:STANDBY?**

Remarks The **PM:AC:STANDBY?** query returns an integer indicating the present inactive time period for putting the instrument in standby mode during AC operation.

Argument	Value	Inactive Time Period
<i>Value</i>	0	1 min
	1	5 min
	2	30 min
	3	2 hours
	4	NEVER

Related Commands: **PM:AC:STANDBY**; **PM:AC:LEVEL**

8.3.3.14 **PM:ANALOGFILTER**

Description Analog filter select command

Syntax **PM:ANALOGFILTER** *value*

Remarks The **PM:ANALOGFILTER** command selects the analog filter setting.

Argument	Value	Analog Filter
<i>Value</i>	0	None
	1	250 kHz
	2	12.5 kHz
	3	1 kHz
	4	0.5 Hz

Related Commands: **PM:ANALOGFILTER?**

8.3.3.15 **PM: ANALOGFILTER?**

Description Analog filter query

Syntax **PM:ANALOGFILTER?**

Remarks The **PM:ANALOGFILTER?** query returns an integer indicating the present analog filter setting.

Argument	Value	Analog Filter
<i>Value</i>	0	None
	1	250 kHz
	2	12.5 kHz
	3	1 kHz
	4	0.5 Hz

Related Commands: **PM:ANALOGFILTER**

8.3.3.16 PM:ANALOG:IMP

Description	Analog output impedance select command	
Syntax	PM:ANALOG:IMP <i>value</i>	
Remarks	The PM:ANALOG:IMP command selects the analog output impedance.	
Argument	Value	Output Impedance
<i>Value</i>	0	50 Ω
	1	100 k Ω
	2	1 M Ω
Related Commands: PM:ANALOG:IMP?		

8.3.3.17 PM:ANALOG:IMP?

Description	Analog output impedance query	
Syntax	PM:ANALOG:IMP?	
Remarks	The PM:ANALOG:IMP? query returns an integer indicating the present analog output impedance.	
Argument	Value	Output Impedance
<i>Value</i>	0	50 Ω
	1	100 k Ω
	2	1 M Ω
Related Commands: PM:ANALOG:IMP		

8.3.3.18 PM:ANALOG:OUT

Description	Analog output range select command	
Syntax	PM:ANALOG:OUT <i>range</i>	
Remarks	The PM:ANALOG:OUT command selects the analog output range.	
Argument	Value	Max. Output
<i>Range</i>	0	1 V
	1	2 V
	2	5 V
	3	Reserved
Related Commands: PM:ANALOG:OUT?		

8.3.3.19 PM:ANALOG:OUT?

Description Analog output range query

Syntax **PM:ANALOG:OUT?**

Remarks The **PM:ANALOG:OUT?** query returns an integer indicating the present analog output range.

Response	Value	Max. Output
<i>Range</i>	0	1 V
	1	2 V
	2	5 V
	3	Reserved

Related Commands: **PM:ANALOG:OUT**

8.3.3.20 PM:ATT

Description Attenuator enable command

Syntax **PM:ATT** *enable*

Remarks Indicates whether or not the attenuator for the 818 Series power detector is on the detector.

Argument	Type	Description
<i>Enable</i>	int	Enable use of detector responsivity with attenuator available in the calibration module for 818 detectors.

Related Commands: **PM:ATT?**

8.3.3.21 PM:ATT?

Description Attenuator enable query

Syntax **PM:ATT?**

Remarks The **PM:ATT?** query returns 1 when using attenuator calibration, 0 when calculating power without attenuator data.

Response	Value	Description
attenuator	0	Calibrating power without attenuator
	1	Calibrating power using attenuator

Related Commands: **PM:ATT**

8.3.3.22 PM:ATTSN?**Description** Attenuator serial number query**Syntax** **PM:ATTSN?****Remarks** The **PM:ATTSN?** query returns the serial number of the attenuator. When no detector is found the Optical Meter responds with "no detector".

Response	Type	Description
serial number	string	Serial number of the attenuator

Related Commands: **PM:DETMODEL?**, **PM:DETSN?****8.3.3.23 PM:AUTO****Description** Auto range enable command**Syntax** **PM:AUTO** *mode***Remarks** The **PM:AUTO** command sets the power ranging to either manual or automatic.

Argument	Value	Description
<i>mode</i>	0	Manual Optical Meter ranging
	1	Automatic Optical Meter ranging

Related Commands: **PM:AUTO?**, **PM:RANge****8.3.3.24 PM:AUTO?****Description** Auto range mode query**Syntax** **PM:AUTO?****Remarks** The **PM:AUTO?** query returns a value to indicate if auto ranging is enabled or not.

Response	Value	Description
<i>mode</i>	0	Manual Optical Meter ranging
	1	Automatic Optical Meter ranging

Related Commands: **PM:AUTO**, **PM:RANG****8.3.3.25 PM:BAT:BackLight****Description** Inactive time period for reducing the backlight level during battery operation**Syntax** **PM:BAT:BackLight** *value***Remarks** If the instrument is on battery power, its display backlight will be reduced to a pre-specified level following a period of inactivity specified using this command. The instrument

is considered to be inactive if either the keys on the front panel are not being pressed or if there is no communication with a host PC over USB interface.

Argument	Value	Inactive Time Period
Value	0	1 min
	1	5 min
	2	30 min
	3	2 hours
	4	Always

Related Commands: **PM:BAT:BackLight?; PM:BAT:LEVEL**

8.3.3.26 PM:BAT:BackLight?

Description Inactive time period for reducing the backlight level during battery operation.

Syntax **PM:BAT:BackLight?**

Remarks The **PM:BAT:BackLight?** query returns an integer indicating the present inactive time period for reducing the backlight level during battery operation.

Argument	Value	Inactive Time Period
Value	0	1 min
	1	5 min
	2	30 min
	3	2 hours
	4	Always

Related Commands: **PM:BAT:BackLight; PM:BAT:LEVEL**

8.3.3.27 PM:BAT:LEVEL

Description Display backlight level following an inactive time period during battery operation

Syntax **PM:BAT:LEVEL** *value*

Remarks If the instrument is on battery power, its display backlight will be reduced following a pre-specified period of inactivity to the level specified using this command. The instrument is considered to be inactive if either the keys on the front panel are not being pressed or if there is no communication with a host PC over USB interface.

Argument	Value	Backlight Level
Value	0	100%
	1	75%
	2	50%
	3	25%

4 OFF

Related Commands: PM:BAT:BackLight?; PM:BAT:LEVEL?**8.3.3.28 PM:BAT:LEVEL?****Description** Display backlight level following an inactive time period during battery operation**Syntax** PM:BAT:LEVEL?**Remarks** The PM:BAT:LEVEL? query returns an integer indicating the display backlight level following an inactive time period during battery operation.

Argument	Value	Backlight Level
<i>Value</i>	0	100%
	1	75%
	2	50%
	3	25%
	4	OFF

Related Commands: PM:BAT:BackLight; PM:BAT:LEVEL**8.3.3.29 PM:BAT:STANDBY****Description** Inactive time period for putting the instrument in standby mode during battery operation**Syntax** PM:BAT:STANDBY *value***Remarks** If the instrument is on battery power, it will be put in standby mode following a period of inactivity specified using this command. The instrument is considered to be inactive if either the keys on the front panel are not being pressed or if there is no communication with a host PC over USB interface.

Argument	Value	Inactive Time Period
<i>Value</i>	0	1 min
	1	5 min
	2	30 min
	3	2 hours
	4	Never

Related Commands: PM:BAT:STANDBY?; PM:BAT:LEVEL?

8.3.3.30 PM:BAT:STANDBY?

Description Inactive time period for putting the instrument in standby mode during battery operation

Syntax **PM:BAT:STANDBY?**

Remarks The **PM:BAT:STANDBY?** query returns an integer indicating the present inactive time period for putting the instrument in standby mode during battery operation.

Argument	Value	Inactive Time Period
<i>Value</i>	0	1 min
	1	5 min
	2	30 min
	3	2 hours
	4	NEVER

Related Commands: **PM:BAT:STANDBY; PM:BAT:LEVEL**

8.3.3.31 PM:CALDATE?

Description Detector calibration date query

Syntax **PM:CALDATE?**

Remarks The **PM:CALDATE?** query returns the calibration date of the detector.

For example: 21JUN1999.

Response	Type	Description
<i>Date</i>	string	DDMMMYYYYY

8.3.3.32 PM:CALTEMP?

Description Temperature at which detector was calibrated query

Syntax **PM:CALTEMP?**

Remarks The **PM:CALTEMP?** query returns the temperature (degrees Celsius) at which the detector was calibrated. When no detector is present, "no detector" is returned.

Response	Type	Description
<i>temperature</i>	float	Returns the temperature at which the detector was calibrated.

8.3.3.33 PM:CORR

Description Power measurement correction settings

Syntax **PM:CORR** *value1*, *value2*, *value3*

Remarks The **PM:CORR** command sets the power measurement correction settings. These settings are used by the power meter to correct the actual power measurement. The corrected power is calculated using the formula provided below:

$$\text{Corrected Measurement} = ((\text{Actual Measurement} * \text{value1}) + \text{value2}) * \text{value3}$$

Argument	Type	Description
<i>value1</i>	float	Multiplier 1 (default = 1.000)
<i>value2</i>	float	Offset (default = 0.000)
<i>value3</i>	float	Multiplier 2 (default = 1.000)

Related Commands: **PM:CORR?**, **PM:Power?**, **PM:DPower?**

8.3.3.34 PM:CORR?

Description Power measurement correction settings query

Syntax **PM:CORR?**

Remarks The **PM:CORR?** command returns the power measurement correction settings. These settings are used by the power meter to correct the actual power measurement. The corrected power is calculated using the formula provided below:

$$\text{Corrected Measurement} = ((\text{Actual Measurement} * \text{value1}) + \text{value2}) * \text{value3}$$

Response	Type	Description
<i>value1</i>	float	Multiplier 1 (default = 1.000)
<i>value2</i>	float	Offset (default = 0.000)
<i>value3</i>	float	Multiplier 2 (default = 1.000)

Related Commands: **PM:CORR**, **PM:Power?**, **PM:DPower?**

8.3.3.35 PM:DETMODEL?

Description Detector model query

Syntax **PM:DETMODEL?**

Remarks The **PM:DETMODEL?** query returns the model number of the detector.

For example: 818-SL.

Response	Type	Description
model	string	Detector model number

8.3.3.36 **PM:DETSIZE?**

Description Detector surface area

Syntax **PM:DETSIZE?**

Remarks The **PM:DETSIZE?** query returns the surface area of the detector in cm².

Response	Type	Description
Area	float	Detector surface area

8.3.3.37 **PM:DETSN?**

Description Detector serial number query

Syntax **PM:DETSN?**

Remarks The **PM:DETSN?** query returns the serial number of the detector.

For example: 0001

Response	Type	Description
serial number	string	Detector serial number

8.3.3.38 **PM:DIGITALFILTER**

Description Digital filter select command

Syntax **PM:DIGITALFILTER** *value*

Remarks The **PM:DIGITALFILTER** command specifies the digital filter setting.

Argument	Type	Description
<i>Value</i>	int	digital filter size between 0 and 10000

Related Commands: **PM:DIGITALFILTER?**

8.3.3.39 **PM:DIGITALFILTER?**

Description Digital filter query

Syntax **PM:DIGITALFILTER?**

Remarks The **PM:DIGITALFILTER?** query returns the present digital filter setting.

Argument	Type	Description
<i>Value</i>	int	digital filter size between 0 and 10000

Related Commands: **PM:DIGITALFILTER**

8.3.3.40 **PM:DPower?**

Description Detector power query

Syntax **PM:DPower?**

Remarks The **PM:DPower?** returns the actual power measurement. This measurement does not include any correction settings specified using “PM:CORR” command. When the correction settings are set to default values, the power measurement returned by this command is same as the measurement returned by “PM:P?” command.

Response	Type	Description
<i>Power</i>	float	Power in Watts

Related Commands: **PM:Power?**

8.3.3.41 **PM:DS:BUfFer**

Description Data Store buffer behavior selection

Syntax **PM:DS:BUfFer** *behavior*

Remarks The **PM:DS:BUfFer** command selects the behavior mode for control of the Data Store buffer.

Argument	Value	Description
<i>Mode</i>	0	Fixed Size
	1	Ring Buffer

The behavior of the ring buffer is to allow continual data collection after the buffer is full where the oldest values will be overwritten when new measurements are taken.

Related Commands: **PM:DS:BUfFer?**

8.3.3.42 **PM:DS:BUfFer?**

Description Data Store buffer behavior selection query

Syntax **PM:DS:BUfFer?**

Remarks The **PM:DS:BUfFer?** query returns the value of the Data Store buffer behavior.

Argument	Type	Description
<i>Behavior</i>	int	See PM:DS:BUFfer for a definition of the behavior status.

8.3.3.43 PM:DS:CLear

Description	Clear the Data Store of all data
Syntax	PM:DS:CLear
Remarks	The PM:DS:CLear command resets the data store to be empty with no values.

Argument	Type	Description
none	-	

8.3.3.44 PM:DS: Count?

Description	Data Store data item count query
Syntax	PM:DS:Count?
Remarks	The PM:DS:Count? query returns the number of measurements that have been collected in the Data Store.

Argument	Type	Description
<i>count</i>	int	The number of measurements that have been collected.

8.3.3.45 PM:DS:ENable

Description	Enable Data Store Collection
Syntax	PM:DS:ENable <i>enable</i>
Remarks	The PM:DS:ENable enables or disables the collection of measurements in the Data Store.

Argument	Value	Description
<i>enable</i>	0	Disabled
	1	Enabled

Data will be collected after the **PM:DS:ENable** command has been called with a parameter of 1. Data collection will stop when the **PM:DS:ENable** command has been called with a parameter of 0 or when a fixed size data buffer is full.

Related Commands: **PM:DS:ENable?**

8.3.3.46 PM:DS:ENable?

Description	Enable Data Store Collection query
Syntax	PM:DS:ENable?

Remarks The **PM:DS:ENable?** query returns the enabled status of the Data Store.

Argument	Type	Description
<i>enable</i>	int	See PM:DS:Enable for a description of the enable argument

8.3.3.47 **PM:DS:GET?**

Description Retrieve Data Store data query

Syntax **PM:DS:GET? *num***

Remarks The **PM:DS:GET?** command returns a number of measurements that have been collected in the Data Store.

Argument	Type	Description
<i>selection</i>	string	“1” – returns the single value specified “1-10” – returns values in the range from 1-10 “-5” – returns the oldest 5 values (same as 1-5) “+5” – returns the newest 5 values

Note: depending on the number of data points requested, there may be several read operations required on the USB interface.

8.3.3.48 **PM:DS:INTerval**

Description Data Store Interval Select

Syntax **PM:DS:INTerval <interval>**

Parameters

The parameter <interval> is of type <number> that is an integer. The parameter represents the interval in milliseconds for storing one measurement in the data buffer.

Function

This command sets the interval in milliseconds to be used for data storing. For example if DSINT = 100 and DSSIZE = 100 it will take 100 x 100ms to fill the buffer.

Related Commands: **PM:DS:SIZE,PM:DS:SIZE?,PM:DS:INTerval?**

8.3.3.49 **PM:DS:INTerval?**

Description Data Store Interval Query

Syntax **PM:DS:INTerval?**

Parameters None

Function

This query returns the interval in milliseconds currently used for data storing.

Related Commands: PM:DS:SIZE,PM:DS:SIZE?,PM:DS:INTerval

8.3.3.50

PM:DS:SAVEBUFFER

Description Save the current Data Store data to a file

Syntax **PM:DS:SAVEBUFFER** *filename*

Remarks The **PM:DS:SAVEBUFFER** command saves the current user Data Store to a file on the WinCE compatible USB Flash Disk plugged into the USB Host port on the front of the Optical Meter.

Argument	Type	Description
<i>Filename</i>	<i>string</i>	The filename that will be created or overwritten on the USB Flash Disk containing the measurments stored in the Data Store.

8.3.3.51

PM:DS:SIZE

Description Size of the Data Store query

Syntax **PM:DS:SIZE** <size>

Parameters

The parameter <size>is of type <integer> in the range 1 to 250000.The parameter represents the size of the data buffer to be used for data storing.

Function

This command sets the size of the buffer used for data storing.

NOTE

The data buffer is cleared automatically when this command is used and all previously stored data will be gone.

Related Commands: PM:DS:SIZE?,PM:DS:INTerval,PM:DS:INTerval?

8.3.3.52

PM:DS:SIZE?

Description Data Store Buffer Size Query

Syntax **PM:DS:SIZE?**

Parameters None

Function This query returns the data store buffer size.

Returns <size> is of type <number>and represents an integer of the range 1 to 250000.

Related Commands: PM:DS:SIZE,PM:DS:INTerval,PM:DS:INTerval?

8.3.3.53 PM:DS:UNITs?**Description** Data store units query**Syntax** **PM:DS:UNITs?****Remarks** The **PM:DS:UNITs?** query returns an integer indicating the units selected.

Response	Value	Description
<i>units</i>	0	Amps
	1	Volts
	2	Watts
	3	Watts/cm ²
	4	Joules
	5	Joules/cm ²
	6	dBm
	7-10	Reserved
	11	Sun

Related Commands: **PM:UNITs,PM:UNITs?**

Note: The UNITs selection depends on the detector used. Refer to Table 1 for the relationship between the detector type and units.

8.3.3.54 PM:FILT**Description** Filter select command**Syntax** **PM:FILT** *filter type***Remarks** The **PM:FILT** command select the filtering operation to be performed on power readings.

Argument	Value	Description
<i>Filter type</i>	0	No filtering
	1	Analog filter
	2	Digital filter
	3	Analog and Digital filter

Related Commands: **PM:FILT?****8.3.3.55 PM:FILT?****Description** Filter type query**Syntax** **PM:FILT?****Remarks** The **PM:FILT?** query returns an integer indicating the present filter mode.

Response	Value	Description
<i>Filter type</i>	0	No filtering
	1	Analog filter

2	Digital filter
3	Analog and Digital filter

Related Commands: **PM:FILT**

8.3.3.56 **PM:Lambda**

Description Wavelength set command

Syntax **PM:Lambda** *value*

Remarks The **PM:Lambda** command selects the wavelength to use when calculating power. The value must fall within the calibrated wavelength of the detector.

Argument	Type	Description
<i>Value</i>	int	Wavelength in nanometers (nm)

Related Commands: **PM:Lambda?**, **PM:MAX:Lambda?**, **PM:MIN:Lambda?**

8.3.3.57 **PM:Lambda?**

Description Wavelength query

Syntax **PM:Lambda?**

Remarks The **PM:Lambda?** query returns the selected wavelength in nanometers. This is the wavelength used to look up the responsivity from the calibration data.

Response	Type	Description
<i>Wavelength</i>	int	Wavelength in nanometers (nm)

Related Commands: **PM:Lambda**, **PM:MAX:Lambda?**, **PM:MIN:Lambda?**

8.3.3.58 **PM:MAX:Lambda?**

Description Maximum wavelength query

Syntax **PM:MAX:Lambda?**

Remarks The **PM:MAX:Lambda?** query returns the longest calibrated wavelength in nanometers. If no detector is preset, the max lambda for the last read detector is returned.

Response	Type	Description
<i>Wavelength</i>	int	Wavelength in nanometers (nm)

Related Commands: **PM:MIN:Lambda?**

8.3.3.59 PM:MAX:Power?

Description	Maximum power query	
Syntax	PM:MAX:Power?	
Remarks	The PM:MAX: Power? returns current range's maximum readable power.	
Response	Type	Description
<i>Power</i>	float	Power in Watts

8.3.3.60 PM:MIN:Lambda?

Description	Minimum wavelength query	
Syntax	PM:MIN:Lambda?	
Remarks	The PM:MIN:Lambda? query returns the shortest calibrated wavelength in nanometers. If no detector is preset,min lambda for the last read detector is returned.	
Response	Type	Description
<i>Wavelength</i>	int	Wavelength in nanometers (nm)
Related Commands: PM:MAX:Lambda?		

8.3.3.61 PM:MIN:Power?

Description	Minimum power query	
Syntax	PM:MIN:Power?	
Remarks	The PM:MIN: Power? returns current range's minimum readable power.	
Response	Type	Description
<i>Power</i>	float	Power in Watts
Related Commands: PM:MAX:Power?		

8.3.3.62 PM:MEAS:TIMEOUT

Description	Measurement timeout set command	
Syntax	PM:MEAS:TIMEOUT <i>value</i>	
Remarks	This command sets the measurement timeout period. This value is used for making the following measurements: 1. Peak-to-peak measurements: These measurements are updated once every timeout period when the power meter is in Peak-Peak Continuous mode.	

2. Auto-ranging in Pulse Continuous mode: The power meter automatically shifts to a lower range once every timeout period when it determines that no pulse measurements could be taken in the existing range. Users must set this timeout value to 250ms or larger than their pulse repetition rate in order to be able to perform measurements accurately.

Argument	Type	Description
<i>Value</i>	int	Timeout value (milliseconds)

Related Commands: PM:MEAS:TIMEOUT?

8.3.3.63 PM:MEAS:TIMEOUT?

Description	Measurement timeout query
Syntax	PM:MEAS:TIMEOUT?
Remarks	The PM:MEAS:TIMEOUT? query returns the selected measurement timeout value in milliseconds.

Response	Type	Description
<i>Timeout</i>	int	measurement timeout in milliseconds

Related Commands: PM:MEAS:TIMEOUT

8.3.3.64 PM:MODE

Description	Acquisition mode select
Syntax	PM:MODE <i>mode</i>
Remarks	The PM:MODE command selects the acquisition mode for acquiring subsequent readings.

Argument	Value	Description
<i>Mode</i>	0	DC Continuous
	1	DC Single
	2	Integrate
	3	Peak-to-peak Continuous
	4	Peak-to-peak Single
	5	Pulse Continuous
	6	Pulse Single
	7	RMS

Related Commands: PM:MODE?

Note: The MODE selection depends on the detector used. Refer to Table 1 for the relationship between the detector type and MODE.

8.3.3.65 PM:MODE?**Description** Acquisition mode query**Syntax** **M:MODE?****Remarks** The **PM:MODE?** query returns an integer indicating the present acquisition mode.

Response	Value	Description
<i>Mode</i>	0	DC Continuous
	1	DC Single
	2	Integrate
	3	Peak-to-peak Continuous
	4	Peak-to-peak Single
	5	Pulse Continuous
	6	Pulse Single
	7	RMS

Related Commands: **PM:MODE**

Note: The UNITS selection depends on the detector used. Refer to Table 1 for the relationship between the detector type and units.

8.3.3.66 PM:Power?**Description** Power query**Syntax** **PM:P?****Remarks** The **PM:P?** query returns the power in the selected units.**Response Type** Description *power exp* Exponential form (i.e. 9.4689E-04)**Related Commands:** **PM:UNITS?, PM:PWS?****8.3.3.67 PM:PWS?****Description** Read with status query**Syntax** **PM:PWS?****Remarks** The **PM:PWS?** query returns two values that are comma delimited: The first value is the power reading and the second value is the status. The status should be used to validate the reading. The reading is in units corresponding to the units field in the status word.

Response	Type	Description
<i>Power reading</i>	exp	Power in present units
<i>Status</i>	int	A bitfield in hexadecimal defining the current channel status

NOTE

The bitfield is defined as follows:

Bits 10-7	Channel Units. See PM:UNITS?
Bits 6-4	Channel Range, See PM:RANgE?
Bit 3	Detector Present
Bit 2	Channel range change status. Indicates if a measurement has been taken while the unit is ranging
Bit 1	Detector Saturated (reserved, follows bit 0)
Bit 0	Channel overrange. Indicates that the current measurement is overrange for the current channel range

Related Commands: PM:MODE?
Example

If the query returns the following values

1.862153E-004, 118

then the Power reading is 1.862153E-004 and the Status is the hex value 118 or the binary value 00100011000b. Taking into consideration the bitfield definition, the status is decoded as follows:

Bits	Binary	Decimal	Function
10 - 7	0010	2	units = Watts
6 - 4	001	1	range # 1
3	1	1	detector present
2	0	0	instrument not ranging during measurement
1	-	-	reserved
0	0	0	measurement is not over-range

8.3.3.68
PM:RANgE

Description Range select

Syntax **PM:RANgE** *range*

Remarks The **PM:RANgE** command selects the gain stage when making readings from the detector head. The range of this value depends on the detector being used.

Response	Type	Description
<i>range</i>	int	Values range from 0 to 5 for Thermopile and Energy detectors or 0 to 4 for Photodiodes (see Section 2.3.2).

Related Commands: **PM:RANgE?**, **PM:AUTO**

8.3.3.69 PM:RANge?**Description** Range query**Syntax** **PM:RANge?****Remarks** The PM:RANge? query returns an integer that indicates the present range. The range of this value depends on the detector being used.

Response	Value	Description
<i>range</i>	0	Values range from 0 to 5 for Thermopile and Energy detectors or 0 to 4 for Photodiodes (see Section 2.3.2).

Related Commands: **PM:RANge, PM:AUTO****8.3.3.70 PM:REF:VALue****Description** Reference Value Define**Syntax** **PM:REF:VALue <val>****Parameters**

The parameter <val> is of type <number>.

Function

This command provides a means of directly storing a reference value to be used in linear and logarithmic (dB) relative measurements. The units of this value are the current units being used by the instrument.

Related Commands: **PM:REF:STOre, PM:REF:VALue?****8.3.3.71 PM:REF:VALue?****Description** Reference Value Query**Syntax** **PM:REF:VALue?****Parameters** None**Function**

This query returns the user defined reference value. The units of this value are the current units being used by the instrument.

Returns <refval>

<refval> is of type <number>

Related Commands: **PM:REF:STOre, PM:REF:VALue****8.3.3.72 PM:REF:STOre****Description** Reference Value Store

Syntax **PM:REF:STOre**

Parameters None

Function

This command takes the latest reading and stores it as a reference reading to be used when making relative linear and dB measurements. The units of this value are the current units being used by the instrument.

Related Commands: **PM:REF:VALue,PM:REF:VALue?**

8.3.3.73 PM:RESPonsivity?

Description Responsivity query

Syntax **PM:RESP?**

Remarks The **PM:RESP?** query returns the responsivity currently used in making power calculations.

Response	Type	Description
<i>Responsivity</i>	float	Optical Meter responsivity

Related Commands: **PM:UNITS?, PM:PWS?**

8.3.3.74 PM:RUN

Description Run command

Syntax **PM:RUN** *mode*

Remarks The **PM:RUN** command disables or enables the acquisition of data.

Argument	Value	Description
<i>Mode</i>	0	Stop
	1	Run

Related Commands: **PM:RUN?, PM:MODE?**

8.3.3.75 PM:RUN?

Description Run query

Syntax **PM:RUN?**

Remarks The **PM:RUN?** query returns an integer indicating the present run mode.

Response	Value	Description
<i>Mode</i>	0	Stopped
	1	Running

Related Commands: **PM:RUN, PM:MODE?**

8.3.3.76 PM:SATLEVEL

Description Set detector saturation current density or power level

Syntax **PM:SATLEVEL** *value*

Remarks This command sets or overrides the saturation density value recorded in the detector calibration

Argument	Type	Description
<i>Value</i>	float	saturation current density (A/cm ²) for semiconductor detector or saturation power (W) for thermopile detector.

Related Commands: **PM:SPOTSIZE**, **PM:SATLEVEL?**

8.3.3.77 PM:SATLEVEL?

Description Query detector saturation current density or power level

Syntax **PM:SATLEVEL?**

Remarks This command query the saturation density value which is set by PM:SATLEVEL command

Response	Type	Description
<i>Value</i>	float	saturation current density (A/cm ²) for semiconductor detector or saturation power (W) for thermopile detector.

Related Commands: **PM:SPOTSIZE**, **PM:SATLEVEL**

8.3.3.78 PM:SPOTSIZE

Description Set detector spot size

Syntax **PM:SPOTSIZE** *value*

Remarks This command sets the detector spot size. By default, the spot size is same as a detector's surface area. This value is used to when measurement units are set to W/cm², J/cm² or Sun.

Argument	Type	Description
<i>Value</i>	float	Spot size (cm ²)

Related Commands: **PM:SPOTSIZE?**, **PM:DETSIZE?**

8.3.3.79 PM:SPOTSIZE?

Description Detector spot size query

Syntax **PM:SPOTSIZE?**

Remarks The **PM:SPOTSIZE?** query returns the detector spot size in cm².

Response	Type	Description
<i>Spot size</i>	float	detector spot size in cm ²

Related Commands: **PM:SPOTSIZE**, **PM:DETSIZE?**

8.3.3.80

PM:STAT:MAX?

Description Statistics Maximum Value Query

Syntax **PM:STAT:MAX?**

Parameters None

Function This query returns the maximum value in the statistics buffer.

Returns <max>
<max>is of type <number>in exponent notation.

8.3.3.81

PM:STAT:MEAN?

Description Statistics Mean Value Query

Syntax **PM:STAT:MEAN?**

Parameters None

Function This query returns the mean or average of all the values in the statistics buffer.

Returns <mean>
<mean>is of type <number>in exponent notation.

8.3.3.82

PM:STAT:MIN?

Description Statistics Minimum Value Query

Syntax **PM:STAT:MIN?**

Parameters None

Function This query returns the minimum value in the statistics buffer.

Returns <min>
<min>is of type <number>in exponent notation.

8.3.3.83

PM:STAT:MAXMIN?

Description Statistics Max-Min Query

Syntax **PM:STAT:MAXMIN?**

Parameters	None
Function	This query returns the difference between the maximum and minimum readings in the statistics buffer.
Returns	<mxmn> <mxmn>is of type <number>in exponent notation.

8.3.3.84 PM:STAT:SDEVIation?

Description	Statistics Standard Deviation Query
Syntax	PM:STAT:SDEVIation?
Parameters	None
Function	This query returns the standard deviation of the readings in the statistics buffer.
Returns	<stddev> <stddev>is of type <number>in exponent notation.

8.3.3.85 PM:Temp?

Description	918 detectors temperature query	
Syntax	PM:Temp?	
Remarks	The PM:Temp? query returns the 918 detector's temperature as a float in degrees Celsius.	
Response	Type	Description
	temp	float Detector temperature in degrees Celsius (°C)
Related Commands: PM:ATT?		

8.3.3.86 PM:THERM:PREDICT

Description	Thermopile prediction command	
Syntax	PM:THERM:PREDICT <i>mode</i>	
Remarks	The PM:THERM:PREDICT command disables or enables the thermopile prediction algorithm. When this algorithm is enabled, the instrument predicts the power measured by thermopile detector very accurately, a few seconds faster than the natural response of the detector.	
Argument	Value	Description
	0	Disable prediction algorithm
	1	Enable prediction algorithm
Related Commands: PM:THERM:PREDICT?		

8.3.3.87 PM:THERM:PREDICT?**Description** Thermopile prediction query**Syntax** **PM:THERM:PREDICT?****Remarks** The **PM:THERM:PREDICT?** query returns the state of thermopile prediction algorithm.

Response	Value	Description
<i>Mode</i>	0	Prediction algorithm is disabled
	1	Prediction algorithm is enabled

Related Commands: **PM:THERM:PREDICT****8.3.3.88 PM:TRIG:START****Description** This command sets the optional start event.**Syntax** **PM:TRIG:START** *option***Parameters**

The parameter <option> is of type <number> and is an integer from 0 to 3.

Argument	Value	Description
<i>option</i>	0	Continuous measurement
	1	Reserved
	2	Measurement starts when a designated Soft key is pressed
	3	Measurement starts when PM:TRIG:STATE 1 command is issued.

Related Commands: **PM:TRIG: START?****8.3.3.89 PM:TRIG:START?****Description** Trigger START Query**Syntax** **PM:TRIG:START?****Parameters** None**Function**

This query returns the TRIGGER START condition.

Related Commands: **PM:TRIG: START****8.3.3.90 PM:TRIG:STOP****Description** This command sets the optional stop event.

Syntax **PM:TRIG:STOP** <option>

Parameters

The parameter <option> is of type <number> and is an integer from 0 to 6.

Argument	Value	Description
<i>option</i>	0	The measurement never stops
	1	Reserved
	2	Measurement stops when a designated Soft key is pressed
	3	Measurement stops when PM:TRIG:STATE 0 command is issued.

Related Commands: PM:TRIG: STOP?

8.3.3.91 PM:TRIG:STOP?

Description Trigger STOP Query

Syntax **PM:TRIG:STOP?**

Parameters None

Function

This query returns the TRIGGER STOP condition.

Related Commands: PM:TRIG: STOP

8.3.3.92 PM:TRIG:STATE

Description This command sets the trigger state

Syntax **PM:TRIG:STATE** <option>

Parameters

The parameter <option> is of type <number> and is an integer 0 or 1.

Argument	Value	Description
<i>option</i>	0	Trigger is armed. The system waits for a trigger start event to occur in order to make a measurement.
	1	System is triggered and at least one measurement occurred.

Related Commands: PM:TRIG: STATE?

8.3.3.93 PM:TRIG:STATE?

Description Trigger STATE Query

Syntax **PM:TRIG:STATE?**

Parameters None

Function

This query returns the TRIGGER STATE condition.

Related Commands: PM:TRIG: STATE

8.3.3.94

PM:UNITs

Description Units select

Syntax **PM:UNITs** *units*

Remarks The **PM:UNITs** command selects the units for readings.

Argument	Value	Description
<i>units</i>	0	Amps
	1	Volts
	2	Watts
	3	Watts/cm ²
	4	Joules
	5	Joules/cm ²
	6	dBm
	7-10	Reserved
	11	Sun

Related Commands: PM:MODE?

Note: The UNITs selection depends on the detector used. Refer to Table 1 for the relationship between the detector type and units.

8.3.3.95

PM:UNITs?

Description Units query

Syntax **PM:UNITs?**

Remarks The **PM:UNITs?** query returns an integer indicating the units selected.

Response	Value	Description
<i>units</i>	0	Amps
	1	Volts
	2	Watts
	3	Watts/cm ²
	4	Joules
	5	Joules/cm ²
	6	dBm
	7-10	Reserved
	11	Sun

Related Commands: PM:UNITS

Note: The UNITS selection depends on the detector used. Refer to Table 1 for the relationship between the detector type and units.

8.3.3.96 PM:ZEROSTOre

Description Zero value set command

Syntax **PM:ZEROSTO**

Remarks The **PM:ZEROSTO** command sets the zeroing value with the present reading.

Argument	Type	Description
none	-	

Related Commands: PM:ZEROVAL?

8.3.3.97 PM:ZEROVALue

Description Zero value set command

Syntax **PM:ZEROVAL** *value*

Remarks The **PM:ZEROVAL** command sets the zeroing value. When enabled, zeroing subtracts the stored zero value from readings before making measurement calculations.

Argument	Type	Description
<i>Value</i>	float	Zeroing value

Related Commands: PM:ZEROSTO

8.3.3.98 PM:ZEROVALue?

Description Zero value query

Syntax **PM:ZEROVAL?**

Remarks The **PM:ZEROVAL?** query returns the zero value.

Response	Type	Description
<i>value</i>	float	Zero exponent

Related Commands: PM:ZEROVAL

9 Principles of Operation

9.1 Introduction

The 1918-R Optical Meter electronics adapt to a number of signal measurement tasks: DC current or voltage, AC peak-to-peak current or pulse voltage, or integrated DC current or voltage signals. This versatility is required to handle the various signals that Newport's **Low Power, High-Power, Energy** and other detector families generate. These detector families are based on semiconductor, thermopile and pyroelectric. The detector data is introduced to the 1918-R by way of a calibration module specific to the detector in use. At power up (and RESET), the 1918-R downloads information about the detector from the calibration module or the detector internal memory. Based on the calibration module preprogrammed data, the meter learns the set of operating states available to the detector. The user then selects among the available operating states when using the meter. Front panel control and the operating states of the 1918-R Optical Meter are discussed in Section 3.

9.2 Analog Signal Flow

The detector signals can follow many different paths through the 1918-R input amplifier chain. A block diagram of analog signal flow is shown in Figure 58. The actual flow path depends upon the detector type and the mode of measurement.

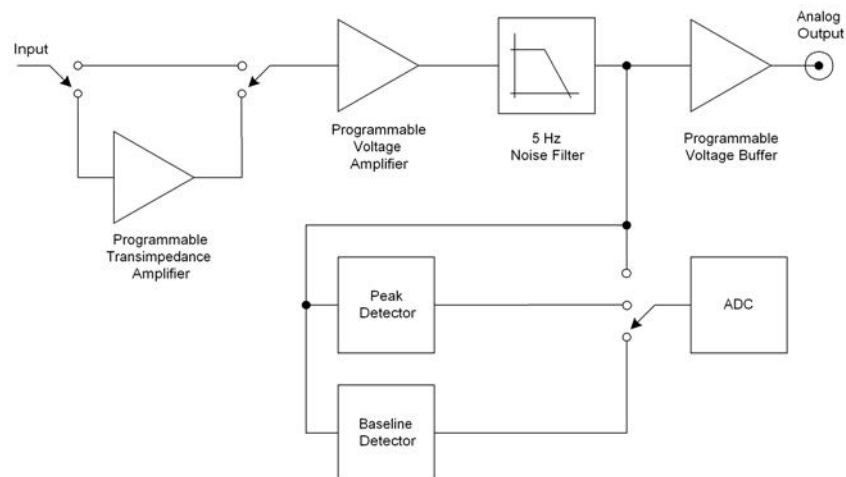


Figure 58 1918-R Optical Meter Analog Signal Flow Diagram

The analog signal flow path is primarily determined by the responsivity units of the detector. The numerator of these units indicates how the meter must be configured in order to obtain a calibrated optical measurement. Analog signal flow is independent of whether single or continuous measurements are made.

Responsivity units and signal flows for the various detector families are listed in Table 4.

Detector Family	Resp. Units	Mode	Amplifier Path	Peak-Baseline
Low-Power	A / W	DC	I	No
Low-Power	A / W	Peak-Peak	I	No
Low-Power	A / W	Integral	I	No
High-Power	V / W	DC	V	No
High-Power	V / W	Integral	V	No
Energy	V / J	Pulse	V	Yes

Table 4 Analog Signal Flow Paths.

9.3 Digitized Signal Flow

An analog-to-digital converter captures the input signal. The input signal has the units of current or voltage depending upon how the input amplifier chain was configured. Note: In this manual the digitized signal may be referred as *sampled signal* or *samples*.

This digitized signal moves through a number of process steps that may or may not alter the digitized value depending upon the operating state of the meter. Each of these possible-processing steps is discussed further.

Digital Filter

If the digital filter annunciator Dig or Digital is on, Section 4.2.5, the filter output is the average of the most recent 10, 100, 1000, 10000 samples. The number of samples is selectable by the user with the Filter menu. Also, the user can change the number of samples. When less than the selected number of samples has been acquired since the last reset of the digital filter, the output is the average of all the values received. The digital filter is reset when the 1918-R Optical Meter is turned on and whenever the UNITS or the range changes or when the MODE of operation changes.

NOTE

When using the digital filter in CW Single acquisition mode, each measurement is the average of the last 10, 100, 1000, 10000 acquisitions independent of how old any of the measurements are.

Gain

Gain processing accounts for the signal gain of the input amplifiers. The output is the product of the digitized value and the amplifier gain.

Zero Offset

Zero offset is active whenever the Offset annunciator is lit. The zero offset output is equal to the input value less the zero reference value.

Responsivity Map(s)

This process scales the input value in accordance with current calibration wavelength and the responsivity map downloaded from the detector calibration module. The output of this process, i.e. the measurement value, is the digitized input value divided by the responsivity associated with the current calibration wavelength or the user defined calibration value. Different responsivities are used depending upon if the attenuator is on as shown in the Setup Menu.

Units Correction

Unit's correction adjusts a measurement value to account for the display units selected. When the display units are equal to the detector signal units, i.e. equal to the numerator of the responsivity units, Table 1, the measurement value is not adjusted. Otherwise the digitized value is adjusted to account for detector responsivity and/or additional unit conversions such as W-to-W/cm².

NOTE

Per area unit conversions such as W-to-W/cm² divide the measurement value by the active area of the detector. This calculation assumes that the entire active area of the detector is uniformly illuminated. Per area measurements where the entire detector active area is not uniformly lit are not accurate. The user must insure that these conditions are met before utilizing per area units or make measurement corrections accordingly.

9.4 Typical Detector Signals

The flexibility of the 1918-R Optical Meter analog signal flow is required in order that it may properly measure the signals that various types of detectors make.

Basic optical power or energy measurements are related to a measured detector signal, S, in the following way:

$$P \text{ or } E = \frac{S}{R_{\lambda}}$$

Where: R_{λ} = Detector responsivity at λ
S = Detector signal

Semiconductor (918D/818 Series Low-Power) detectors provide a current signal. The 1918-R is capable of 10 pA resolution in order to provide the highest sensitivity performance with these detectors.

Thermopile (818P Series High-Power) detectors, provide a small voltage signal.

The 1918-R is capable of 76.3 nV resolution in order to reach the sensitivity limits of thermopile detectors.

Pyroelectric (818E Series Energy) detectors, deliver a peak voltage signal.

The 1918-R is capable of capturing 2 μ sec rise time voltage spikes so that it may be operated with the fastest of these detectors.

9.5 Thermopile Detector Signals

Thermopile detectors respond with a voltage signal that slowly changes in incident optical power. The time constant of most thermopile detectors is on the order of 1 to 10 seconds.

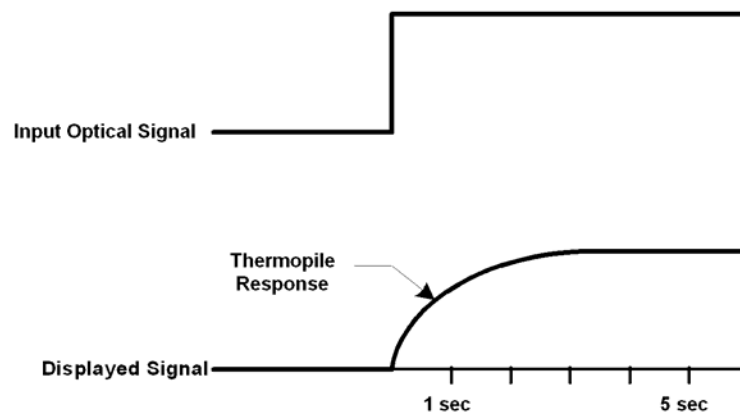


Figure 59 Thermopile Signals exhibit 1 to 10 second time constants.

9.6 Pulse Energy Detector Signals

A Newport Energy detector will respond to a single radiant energy pulse with a voltage pulse at its BNC output. This pulse exhibits a sharp voltage rise to a peak followed by slower voltage decay that “undershoots” zero volts before settling back to zero volts. When a detector is operated within its proper limits, the voltage difference from immediately before the sharp rise to the peak is linearly proportional to the radiant energy.

If a second pulse arrives before the “undershoot” rises back to zero volts, the voltage rise from this pulse will start from an initial negative value. At sufficient energy pulse repetition rates, a negative “baseline” voltage will develop from which the voltage rise must now be measured to achieve accurate energy readings. The 1918-R contains baseline capture circuitry that maintains its accuracy specifications over rep-rates ranging from single pulse to 10 kHz.

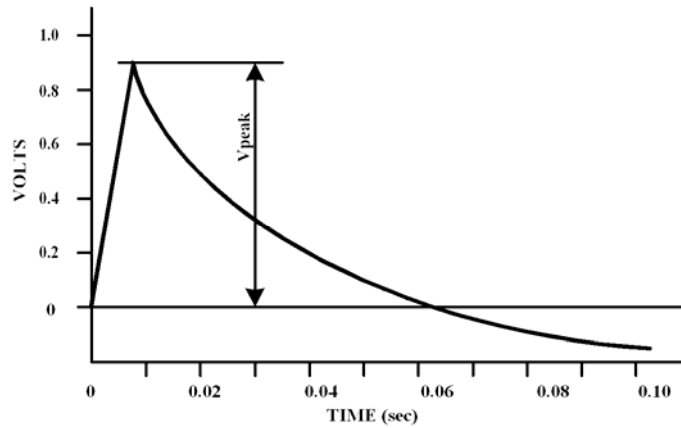


Figure 60 Typical Newport Energy Detector Signal Waveform
 An energy detector signal sharply rises to a peak value and then decays going somewhat negative before finally returning to zero. The energy in the radiant pulse is proportional to the height of the peak measured from immediately before the sharp rise.

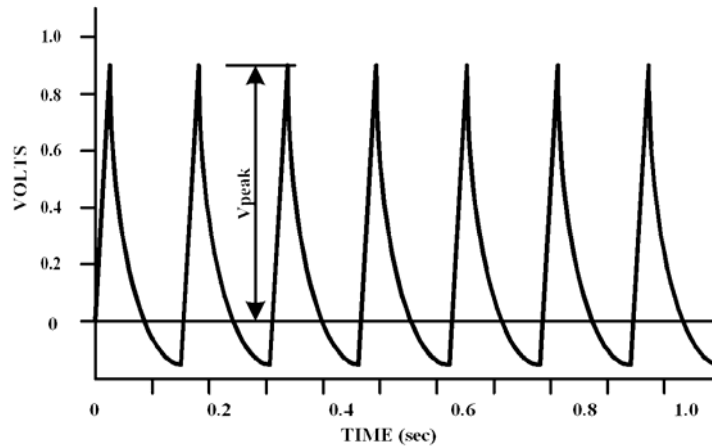


Figure 61 Negative Baseline Voltage.
 Negative Baseline Voltage Due to a Pulse Train shown. If a laser pulse arrives before the previous Energy detector signal has fully decayed, the detector signal rises from the present decay point of the previous signal.

9.7 Peak-to-Peak (Photodiode) Detector Signals

The 1918-R enables one to make peak-to-peak measurements of time varying signals from semiconductor photodiode detectors. Since optical power is a zero bounded positive quantity, signals from a detector observing such modulated light will similarly be zero bounded positive signals. To make a peak-to-peak measurement, the 1918-R must be able to capture both the maximum and minimum values of a detector signal. This is accomplished by sampling the signal.

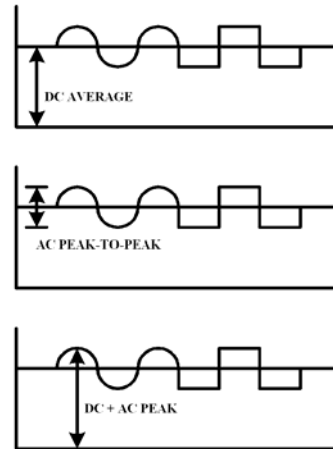


Figure 62 Time Varying Signal Measurements.

Many different measurements can be made on different portions of a time varying signal. The most common are: DC power, peak power, and peak-to-peak power.

9.8 Integration of Detector Signals

The 1918-R provides for making measurements that integrate incoming power detector signals to obtain energy via the CW Integrate mode. In CW Integrate mode, the display units indicate Joules since energy is the time integral of power:

$$E(\lambda) = \int_{t_0}^{t_1} P(\lambda) dt = \int_{t_0}^{t_1} \frac{S(t)}{R(\lambda)} dt$$

As the detector signal actually consists of a stream of digitized values, the integral becomes a numerical approximation using the trapezoid method. See Figure 63.

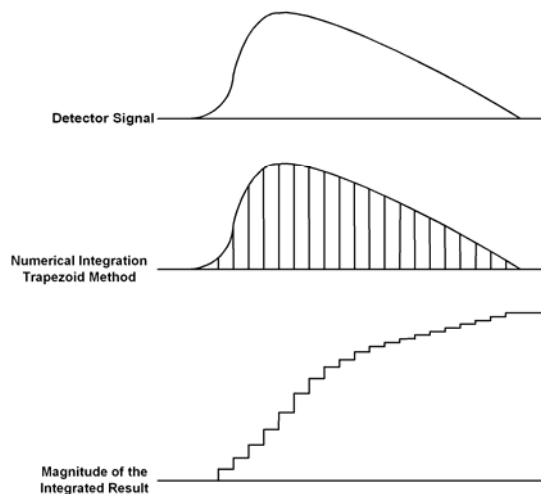


Figure 63 Integrated Energy Via a Trapezoid Approximation.

The INTG measurement mode performs a discrete integration at a 400 Hz sample rate.

Two common applications are natural extensions of the CW Integrate measurement mode:

4. Pulse laser energy measurement using a thermopile detector
5. Energy from exposure over a period of time (dosage)

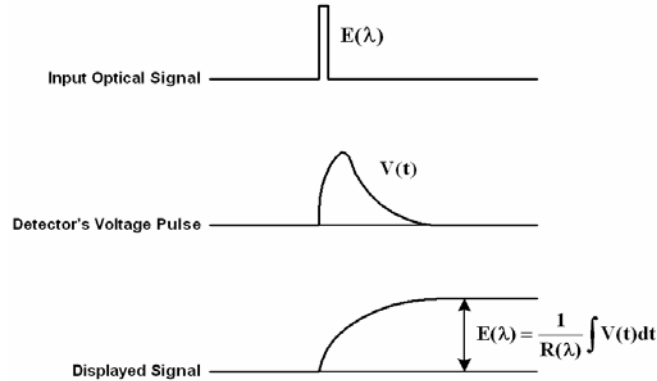


Figure 64 Measuring Laser Pulse Energy with a Thermopile.
Thermopiles are often used to measure pulsed laser energy by integrating the response of the detector to the pulse.

9.9 Analog Output

The 1918-R provides an analog output for signal monitoring. The analog output is the actual amplified detector signal and is uncorrected for the effects of the detector's responsivity, calibration, and ZERO. The analog output signal is a representation of what the meter "sees" at its input. However, the analog output signal may be filtered if the user selects the 0.5 HZ filter (see paragraph 4.2.5). Also, the analog output signal is amplified and scaled to the selected analog output level (see Table 5).

The user can select the analog output level with a command, PM::ANALOG:OUT (see Section 8.3). The analog output maximum level corresponds to the maximum full-scale input range the meter is set on. For example, if one selects the analog output level at 1 V, and the meter is set on 250 mW maximum full-scale input range, the analog output signal will have a transfer function of 250 mW/1 V. Therefore, if the analog output is connected to an oscilloscope or voltmeter and the user reads 0.125 V, the detector power is 125 mW.

The selectable analog output levels are given in Table 5.

Output Range	0	1	2
Full Scale Voltage (Load $\geq 1 \text{ M}\Omega$)	1V	2V	5V

Table 5 Analog Output Range Table.

9.10 Measurement Considerations

This section describes detector characteristics, optical and electrical considerations, and environmental influences on optical measurements. In general, measurement accuracy is limited by the accuracy of the detector calibration.

Accurate measurements, however, are also dependent upon proper set-up, control of temperature and illumination conditions and understanding the factors that affect optical measurements.

9.10.1 Detector Calibration and Uncertainty

Newport Corporation calibrates its detectors using secondary standards directly traceable to the United States National Institute of Science and Technology (NIST), to Great Britain's National Physical Laboratory (NPL), or to National Research Council (NRC) of Canada.

The details and uncertainty of the calibration procedure vary with each detector model but a detailed description of the calibration results is supplied with each individually calibrated detector. In general, detector calibration uncertainty varies from 1% to 8% and varies with wavelength. Each detector will also have some variation in response over its surface. Therefore, for the most reproducible measurements, light should illuminate the detector as uniformly as possible over 80% of the detector's active area.

	CAUTION Avoid focusing a light source onto the detector surface. Inaccurate readings and possible detector damage may result. Consult the detector manual for saturation or damage thresholds.
---	--

NIST trace ability requires that detectors be recalibrated on one-year intervals. As individual detector responses change with time, especially in the ultraviolet, recalibration is necessary to assure confidence in the uncertainty of the measurement. For the most reproducible measurements, the same detector should always be used for measurements that are to be directly compared.

9.10.2 Quantum Detector Temperature Effects

Semiconductor (Newport Low-Power) detectors are affected by temperature. At long wavelengths, quantum detectors typically lose sensitivity with increasing temperature. Additionally, detector dark current increases exponentially with temperature. For silicon detectors, dark current is generally on the order of a few picoamps at room temperature. With uncooled germanium detectors, however, this dark current is on the order of a nanoamp, or typically 1,000 to 10,000 times greater than silicon. Observed

dark current is often dominated by the interaction between the detector and a meter's amplifier and is typically larger than the theoretical dark current limit. Silicon detectors are inherently quieter than germanium detectors due to their higher internal resistance and lower capacitance. The noise or drift in the dark current sets a lower bound on the measurement resolution that can be achieved with any given detector. Cooling a detector significantly lowers its dark current and dark current noise.

The observed dark currents can also be zeroed at any moment via the ZERO function. Since dark currents drift with temperature, the ZERO should be adjusted just prior to taking any measurements. If the detector temperature is constant, sensitivity changes and dark current drifts are significantly reduced.

The 1918-R Optical Meter can measure the detector temperature and correct the power readings accordingly. The power correction happens automatically only for those detectors which are equipped with a thermistor (Newport offers detectors with thermistors for temperature compensation. See Newport web-site www.newport.com for more details.)

The thermistor signal is read via the detector DB-15 connector and the displayed power is adjusted continuously depending on the detector temperature. The analog output signal shows the detector signal and is not corrected for temperature.

9.10.3

Thermopile Detector Temperature Effects

Thermopile (Newport High-Power) detectors are significantly affected by temperature fluctuations arising from airflow disturbances. As the detector is a temperature-measuring device, airflow disturbances set a practical lower limit on the power that a detector can measure. In order to get the most out of any thermopile detector, be careful to shield the detector from airflow disturbances. Common sources of disturbance are: air conditioners and people walking past.

NOTE

*"Use **Prediction** to deactivate the power meter acceleration software that provides the "prediction" response. By using advanced algorithms and known properties of the detector, this software allows the 1936-R to provide a very accurate power measurement a few seconds faster than the natural response of a thermopile power detector. It accelerates the natural response by a factor of 5 to 10.*

Turning off the prediction will result in a slower response but it can provide a more stable measured value in a noisy environment. The default is on. Users are advised to turn the prediction off if they use a high-power thermopile at low power levels."

9.10.4 Energy Detector Temperature Effects

Pyroelectric (Newport Energy) detectors are AC coupled devices and thus are not susceptible to temperature induced DC signal offsets or noise floor changes.

One generally does not need to take much precaution with pyroelectric detectors except to make sure that their damage threshold is not exceeded.

9.10.5 Ambient and Stray Light

Ambient and stray light striking the detector should be considered when making a measurement. Ambient light can be distinguished from dark current (or the detector/meter noise floor) by either turning off or blocking the source and covering the detector face with opaque material such as a piece of black rubber.

Using the human hand to cover the detector is not advised because it emits a significant amount of infrared radiation and radiates a temperature significantly different from ambient. With the detector covered, a reading of the dark current may be made. Next, remove the material that is covering the detector and take another reading. The difference is the ambient light level.

The effects of ambient light are greatly reduce when using a fiber-connectorized signal input to the detector. If free-space beam measurements are desired, using an attenuator (**Low-Power** detectors have an OD3 attenuator included) will reduce stray light and often improve the source signal to ambient signal noise level. Wavelength-specific filters, such as optical cutoff, bandpass, or spike filters can also be used if the signal wavelength spectrum permits. Other techniques to reduce stray light include using apertures, placing detector in a box or other housing to shield the surface from light (or air currents when using Newport's **High-Power** disk thermopile detectors) which is not coming from the source, and turning off room and other light.

NOTE

Changes in ambient light levels can occur from such factors as turning room lights on or off, or by moving people or equipment. Remember, if you can see your detector element, then your detector can see the light bouncing off you.

The effects of ambient light are greatly reduced when using a fiber-connectorized signal input to the detector. If free-space beam measurements are desired, using an attenuator will often improve the signal to ambient signal noise level.

Wavelength-specific filters, such as optical cutoff, band pass, or spike filters can be used if the signal wavelength spectrum permits. Other techniques to reduce stray light include using apertures, placing the detector in a box or other housing to shield the surface from light (or air currents) and turning off room and other polluting light sources.

9.10.6 Signal Filtering

The 1918-R Optical Meter offers the user the option to filter the detector signal. There are two programmable filters that can be used individually or together to condition the detector signal: the Analog Filter and the Digital Filter.

The Analog Filter is a hardware based, programmable low-pass filter. The user can select the following settings: 0.5 Hz, 1kHz, 12.5kHz, and 250kHz. This flexibility is offered to the user to improve the measurement precision for different detectors.

It is well known that white noise has large spectrum. The noise level increases with bandwidth. The higher the noise, the higher the minimum signal that can be measured. The quality of the measurement can be linked to the signal-to-noise ratio (SNR): the higher the SNR, the better the measurement precision of very small signals. In order to increase the SNR the user may choose to cut the bandwidth of the power meter, depending on the detector signal.

For example, if the detector is modulated with a sine wave of 800Hz, the user may choose to set the Analog filter to 1kHz to increase the SNR and be able to measure very small signals. Also, cutting the bandwidth at 1kHz does not affect the measurement. However, if the modulation frequency is 800Hz, but the signal is a square, the user may opt to cut the bandwidth higher, at 12.5kHz, to let the signal harmonics be unfiltered and minimize the distortion. Of course, in this case, the noise in the system will increase with the square root of the bandwidth. The user needs to assess the tradeoffs in his measurement, taking advantage of the 1918-R high flexibility in setting the proper filters for the measurement.

Generally, the 0.5 Hz filter is used for CW Cont. measurements. However, the 1918-R Optical Meter can measure the amplitude of AC signals down to 0.01Hz, so the 0.5 Hz filter can be used for any modulating signal below 0.5 Hz.

The Digital Filter is a moving average filter that can be set at 10, 100, 1000, or 10000 measurements. These values can be changed by the user. If the digital filter is set on 10, the average value is computed on 10 measurements, in a First In First Out (FIFO) fashion. The Digital Filter helps the SNR the same way as the Analog Filter does. With both filters on, low level, continuous signals can be measured with a high degree of accuracy.

The Analog Filter is always applied to the entire amplifier chain of the meter. Therefore, the user will see the detector signal filtered at the Analog Output BNC. The Digital Filter is a digital signal-processing filter and is only applied to the calculated measured values that are displayed on the meter screen. While the displayed values may have both filters applied ($A_n + D_g$) the

Analog Output can only have the Analog Filter applied to the signal. Because of that, the user may see a difference in the way the signal is filtered between the displayed values and the Analog Output values.

9.11 Common Measurement Errors

The most common sources of optical measurement error are listed in Table 6 below. Other common errors are discussed in the preceding subsections of Measurement Considerations.

Type of Error	What should be done?
Radiometry	Check that all of the light is actually hitting the detector
Ambient light	Check that any ambient light was ZEROed before the measurement was made.
Wavelength calibration	Check that the calibration factor for the measurement wavelength is properly set.
Detector saturation or damage	Check that the optical power density remains below the detector's saturation or damage

Table 6 Common Measurement Errors

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10 Troubleshooting

10.1 Power Supply Problems

	Problem	Solution
1	The unit does not turn on when battery powered.	1. Use the external power supply to operate the unit and recharge the battery. 2. Verify that the USB-Host interface is not connected to a shorted cable or device. Verify that the battery is properly installed.
2	The unit does not turn on when powered from the external power supply.	1. Verify that the external power supply is connected to AC power. 2. Verify that the external power supply connector is fully mated with the connector on the instrument.
3	The battery does not charge.	Check the PWR LED. 1. If it is blinking, verify that the unit is operating within the correct temperature range. If the battery is extremely discharged, the PWR LED may remain blinking for up to 30 minutes while the battery is “pre-charged.” 2. If it is not lit: 2.1. Verify that the external power supply is connected to AC power. 2.2. Verify that the external power supply connector is fully mated with the connector on the instrument. 2.3. Verify that the battery is properly installed. 3. If it is solid red, the battery is charging.

		Allow the unit to charge for at least three hours. 4. If it is solid green, the battery is fully charged.
4	Short battery life	1. Allow battery to fully charge. 2. Use the various power-saving modes of the instrument. 3. Disconnect any device(s) connected to the USB-Host interface connector.

10.2 Analog Output Problems

	Problem	Solution
1	There is no AC signal at the analog output	Check if the 0.5 Hz Analog Filter is on.

10.3 Questions and Answers

Q. I am trying to update the firmware but my USB flash drive is not working. I don't see any light on the memory stick.

A. The USB flash drive needs to be WinCE compatible. WinCE is the operating system of 1918C power meter.

Q. Is the detector hot swappable?

A. Yes, you can swap the detector without having to turn off the power meter.

Q. Is it possible to write the measurement data into Spreadsheet?

A. Yes, you can plug in a USB flash drive into the power meter and save the data from the Statistics screen.

Q. Is it possible to use a third party detector with the 1918-R Optical Meter?

A. It may be possible for some measurements by purchasing a special adaptor. Contact Newport Corporation for availability and details.

- Q.** I would like to change the configuration back to the factory default setting. How can I do that?
- A.** Turn off the Power Meter. Hold down both “ZERO and “HOLD” buttons and turn on the Power Meter. A message window appears. Choose YES or NO for reset by using the left-right arrow keys and press enter.
- Q.** The display shows some noise and fluctuation when I measure my CW light source. Any suggestions?
- A.** Turn on the analog/digital filter. Check whether the meter and detector calibration is up-to-date.
- Q.** Can I measure peak power or energy of a single pulse with a Newport low power detector (918D, etc.) and 1918-R?
- A.** No. There is no energy or peak power mode measurement with the 1918-R when using a low power detector.

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11 Maintenance and Service



CAUTION

There are no user serviceable parts inside the 1918-R Optical Meter. Work performed by persons not authorized by Newport Corporation will void the warranty.

Calibration accuracy is warranted for a period of 1 year. After 1 year, the unit should be returned to Newport Corporation for recalibration and NIST traceability re-certification.

11.1 Enclosure Cleaning



WARNING

Before cleaning the enclosure of the 1918-R Optical Meter, the external power supply must be disconnected from the unit.

The enclosure should only be cleaned with isopropyl alcohol or a mild soapy water solution applied to a damp lint-free cloth.

11.2 Battery Pack Replacement

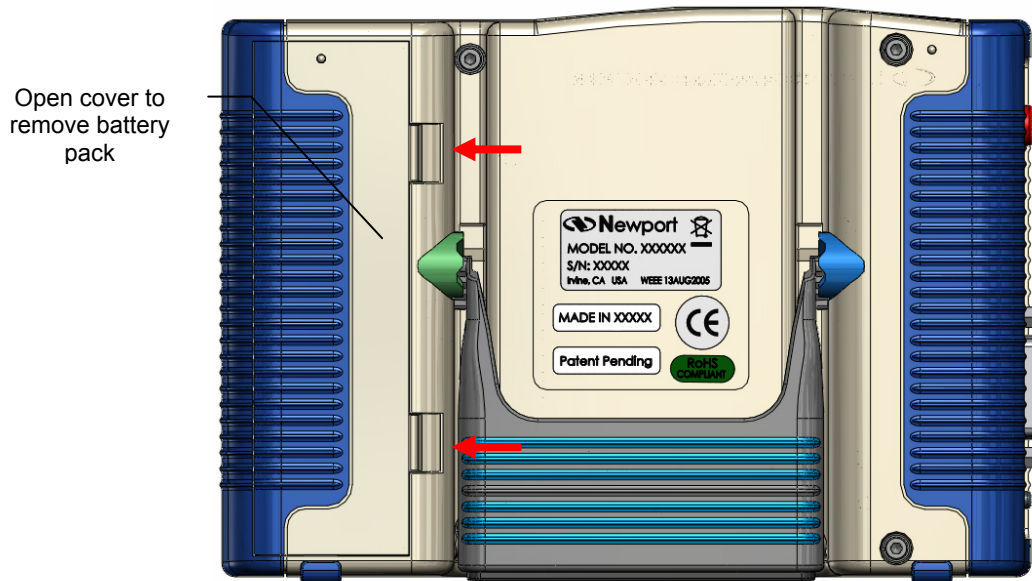


Figure 65 Battery Replacement



WARNING

To reduce the risk of electric shock or damage to the instrument, turn the power switch off and disconnect the power cord before replacing a battery pack.

To replace a battery pack:

1. Disconnect the power cord from the instrument
2. Push the battery cover latches as the arrows show in Figure 65 and lift the battery cover.
3. Disconnect the battery pack from the instrument by detaching the connector from the contacts to which it is attached
4. Remove the old battery pack and replace with a new 1918-BAT battery pack, by attaching the connector to the contacts and placing the battery pack inside the compartment
5. Close the battery cover, while assuring that no wires are caught between the instrument and the back cover.
6. Reconnect the power cord, charge the new battery pack for at least 4 hours and turn on the instrument.

11.3 Spare Parts

The following spare parts can be ordered directly from Newport Corporation.

Item	Newport Part Number	Equivalent
Battery	1918 - BAT	No equivalent. Use only Newport battery.
External Power Supply	1918 – PS	Cincon Electronics # TR36A2423A03
Power Cord	1918 – PSC	Qualtek Electronics #223021-01 or equivalent 8 ft max, SPT-2, 18 AWG, 7A, Nema 1-ISP to IEC 60320K7 cord.
Battery Compartment Cover	90000198	No equivalent.
Kickstand Assembly	45344-01	No equivalent.

Table 7 Spare Parts

11.4 Obtaining Service

The 1918-R Optical Meter contains no user serviceable parts. To obtain information regarding factory service, contact Newport Corporation or your Newport representative. Please have the following information available:

1. Instrument model number (on the rear panel).
2. Instrument serial number (on rear panel).
3. Description of the problem.

If the instrument is to be returned to Newport Corporation, you will be given a Return Number, which you should reference in your shipping documents. Please fill out a copy of the service form, located on the following page, and have the information ready when contacting Newport Corporation. Return the completed service form with the instrument.

11.5 Service Form



Newport®
Experience | Solutions

Newport Corporation
U.S.A. Office: 800-222-6440
FAX: 949/253-1479

Name _____ **Return Authorization #** _____
(Please obtain RA# prior to return of item)

Company _____
(Please obtain RA # prior to return of item)

Address _____ Date _____

Country _____ Phone Number _____

P.O. Number _____ FAX Number _____

Item(s) Being Returned:

Model # _____ Serial # _____

Description _____

Reason for return of goods (please list any specific problems):

12 Appendix A – Syntax and Definitions

12.1 Definition of <string>

For convenience, the 1918-R recognizes double quoted, single quoted, and unquoted strings with certain restrictions as detailed below. Any of these forms may be used where a <string> parameter is required.

- | | |
|----------------------------------|---------------------------|
| 1. <string>,using double quotes | “this is a string” |
| 2. <string>,using single quotes. | ‘this is a string’ |
| 3. <string>,using no quotes. | thisisastring |

A description of each type of <string> follows:

1. <string> defined using double quotes.

A double quote indicates that a string follows, and the string is terminated by another double quote. A double quote may be embedded within the string by using two double quotes together:

Example: “this string contains a “”double quote”

All characters within the two outer double quotes are considered part of the string. It is an error if the string does not terminate with a double quote. The string cannot contain the <CR>(ASCII decimal 13), <LF>(ASCII decimal 10), or End or Identify characters.

2. <string> defined using single quotes.

This form is similar to double quoted string. A single quote indicates that a string follows, and the string is terminated by another single quote. A single quote may be embedded within the string by using two single quotes together:

Example: ‘this string contains a ‘’single quote’

All characters within the two outer single quotes are considered part of the string. It is an error if the string does not terminate with a single quote. The string cannot contain the <CR>(ASCII decimal 13), <LF>(ASCII decimal 10), or End or Identify characters.

3. <string> defined using no quotes.

All strings using this format must start with an alphabetic character (A through Z, a through z). All other characters must be either alphabetic, digit (0 through 9) or the '_' character. Any other character will delimit the string.

Some examples are shown below:

Sent:	this is a string	
Interpreted:	this	(1st string)
	is	(2nd string)
	a	(3rd string)
	string	(4th string)
Sent:	this,isastring	
Interpreted:	this	(1st string)
	,	(separator character)
	isastring	(2nd string)
Sent:	w/cm	
Interpreted:	w	(1st string)
	ERROR	(unrecognized character)
	cm	(2nd string)

12.2 Definition of <number>

The 1918-R recognizes four types as <number>, thus any format may be used.

1. <number> defined as floating point.
2. <number> defined as binary.
3. <number> defined as octal.
4. <number> defined as hexadecimal.

Where necessary, integers are converted to floating point numbers. In all cases, a number is terminated by any of the below characters:

<NL> <EOI> <SPACE>

Any non-valid characters detected in any number received are considered an error in format, and an error condition will be generated in the system.

A description of each type of <number> follows:

1. <number> defined as floating point.

Any of the following characters, as the first character of an ASCII sequence, indicates that a number is being defined:

+-.0 1 2 3 4 5 6 7 8 9

A floating point number is defined as follows:

1. Optional +-sign. This defines the sign of the number. If missing, positive is assumed.
2. Optional 0-9 digits. These digits define the integer portion of the mantissa.
3. Optional . decimal point. This defines the end of the integer portion of the mantissa, and indicates that the fractional portion of the mantissa follows.
4. Optional 0 -9 digits. These digits define the fractional portion of the mantissa.
5. Optional exponent indicator, an ASCII 'E' or 'e', followed by a '+' or '-' (optional), followed by decimal digits.

Examples:

The numbers below all represent the value "1.2 "

1.2

1.2e0

+01.2E+00000

120E-2

.12e1

The numbers below all represent the value "-1.2 "

-1.2

-1.2e+00

-0001.2e+0

-120e-2

.12E1

2. <number>defined as binary.

The 1918-R recognizes unsigned binary numbers in the range of 0 to 65535, decimal, or 0 to 1111111111111111 binary. Binary numbers are represented using only the digits 0 and 1. A binary number has the following format:

#B<*binary*>

Where

#B = mandatory binary number header

<*binary*> = binary digits (0's or 1's)

Example:

All numbers below represent the decimal value 129.

#B10000001

#b010000001

#b10000001

3. *<number>* defined as octal.

The 1918-R recognizes unsigned octal numbers in the range 0 to 65535 decimal, or 0 to 177777 octal. Octal numbers are represented using digits from 0 to 7. An octal number has the following format:

#Q<octal>

Where

#Q = mandatory octal number header

<octal> = octal digits (0 to 7)

Example:

All numbers below represent the decimal value 129.

#Q201

#q0201

#q201

4. *<number>* defined as hexadecimal.

The 1918-R recognizes unsigned hexadecimal numbers in the range 0 to 65535 decimal, or 0 to FFFF hexadecimal. Hexadecimal numbers are represented using the digits 0 -9 and the characters A -F. A hexadecimal number has the following format:

#H<hexadecimal>

Where

#H = mandatory octal number header

<hexadecimal> = hexadecimal digits (0 -9 and A -F)

Example:

All numbers below represent the decimal value 127.

#H7f

#H007F

#h7f

13 **Appendix B – Error Messages**

13.1 Introduction

The communication errors can be retrieved with the following commands: *ERR? or ERRSTR?. The descriptions of the returned errors are detailed in the next sections.

13.2 Command Errors

Command Errors are associated with the conversion of the data received into the commands and their parameters (parsing). Incorrect syntax, incorrect parameters, and improper command format will generate these errors. Any command error will cause the Command Error bit (bit 5) in the Standard Event Status Byte to be set.

104, Numeric Type Not Defined

Generated during the parsing of a number and an undefined number type is encountered.

106, Digit Expected

Generated during the parsing of a number and the parser encounters a non-number when a number is expected.

107, Digit Not Expected

Generated during the parsing of a number and the parser encounters a number when a different character is expected.

115, Identifier Not Valid

Generated when a parameter is not valid or not properly formed.

116, Syntax Error

Occurs when an error in command structure or parameter type is detected. Some typical causes are:

- Using a number as a command mnemonic.
- Using the wrong parameter type.
- Using ASCII characters outside of a string constant that are not defined by the command language syntax.
- Missing or too many parameters.

The above list is not exhaustive but does give the basic idea of what to look for.

126, Too Many Or Few Arguments

Generated when command arguments are missing or too many.

13.3 Execution Errors

Execution Errors are associated with the interpretation of the converted commands and parameters received. Incorrect parameter values and numerical range errors are types of execution errors. Any execution error will cause the Execution Error bit (bit 4) in the Standard Event Status Byte to be set.

1, Out of memory

This error is caused by an internal program fault, and may be followed by an automatic instrument reset.

201, Value Out Of Range

This error will occur if a parameter is out of a valid range or not in the set of valid parameters for a given command.

214, Exceeds Maximum Length

Generated when the command exceeds the maximum command length. Try shortening the command string.

217, No saved information in recalled bin

Generated when the user attempts to recall a bin which has no previous stored data.

301, Query Error

The Query Error occurs when the instrument is in the midst of transmitting a message over a communication bus and the instrument exits remote mode.

303, Input Buffer Overflow

Error generated when the system parser runs out of space during reception of command. It may occur if commands are not terminated correctly. Input buffer is 1,024 characters long.

304, Output Buffer Overflow

Error generated when the system parser runs out of space for query results. It may occur if query results in too much data to be returned in a single response, or if multiple command queries are issued but not read. Output buffer is 4,096 characters long.

305, Parser Buffer Overflow

Error generated when the system parser runs out of space for commands. A command is received into the input buffer then transferred to the parser buffer. This error is generated if the command in the input buffer is too large to fit into the available space in the parser buffer. It is usually generated when commands

are sent to the instrument faster than it can process. Parser buffer is 2,048 characters long.

13.4 Device Errors

Device Errors are associated with some system condition that affects the operation of the meter. Errors associated with data reading will set the appropriate bit but will not generate an error message to avoid jamming the error queue or the interface.

701, Detector Calibration Read or Write Failed.

An error was encountered during a read/write operation to the calibration EPROM in the detector. If problem persists, contact the factory.

703, Optical Meter set to defaults due to Firmware update

After upgrading the firmware, in some cases the changes are significant enough to require resetting the instrument to factory defaults.

704, User reference cannot be changed/stored while you are in units of Watts or dBm. Change to dB or Rel to set the user reference value.

User reference can only be stored in dB or Rel mode.

705, Illegal data store parameter change. Queue cleared.

An action that affects the data store caused the data store queue to be cleared.

706, Digital Filter Interval changed, must not be greater than Data Store interval.

The digital filter interval cannot be greater than the data store interval.

707, Digital Filter Disabled with External Trigger.

Digital filter must be disabled during external trigger mode.

708, There is no new data for a statistics update.

No new data has been stored in the data store since the last statistics update.

709, Statistics are not calculated while Data Store is running.

User attempted to compute statistics while data store was active. Turn off data store and compute.

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14 Appendix C - Legacy Commands Reference

Name	Number of Parameters	Function	Legacy Commands
DISP:BRIGHT	1	Sets the backlight level of the display and the keypad	
DISP:BRIGHT?	NONE	Returns the backlight level of the display and the keypad	
DISP:SCReen	1	Switches to the n - screen display.	
DISP:SCReen?	NONE	Returns displayed screen number, 0 – n/a screen number, 1 – Main, etc.	
PM:ANALOG:OUT	1	Sets the analog output range to the desired level	
PM:ANALOG:OUT?	NONE	Returns the analog output range	
PM:ATT	1	Selects if the attenuator's calibration data is included for power calculation.	ATTN_n
PM:ATT?	NONE	Returns setting if attenuator data should or should not be used when calibrating the Optical Meter.	ATTN_n?
PM:ATTSN?	NONE	Gets the attenuator serial number.	ATTNSN_n?
PM:AUTO	1	Sets the Optical Meter ranging to manual or automatic.	AUTO_n
PM:AUTO?	NONE	Returns 1 if automatic Optical Meter ranging is selected.	AUTO_n?
PM:CALDATE?	NONE	Returns the calibration date of the detector.	CALDATE_n?
PM:CALTEMP?	NONE	Returns the temperature at which the calibration was performed.	CALTEMP_n?
PM:CHANnel	1	Selects the Optical Meter channel to display and control.	
PM:CHANnel?	NONE	Returns the Optical Meter channel currently selected.	
PM:DETMODEL?	NONE	Returns the model number of the detector.	DETMODEL_n?
PM:DETSN?	NONE	Returns the serial number of the detector.	DETSN_n?
PM:DS:BUfFer	1	Set data store behavior select.	DSBUF_n
PM:DS:BUfFer?	NONE	Return data store behavior select.	DSBUF_n?
PM:DS:CLear	NONE	Clear data store.	DSCLR_n
PM:DS:Count?	NONE	Return data store count of items stored.	DSCNT_n?
PM:DS:ENable	1	Set data store enable.	DSE_n
PM:DS:ENable?	NONE	Return data store enable.	DSE_n?
PM:DS:GET?	1	Return data store data. {1,1-10,-5,+5} – value, range, oldest 5, newest 5	DS_n?
PM:DS:SIZE	1	Sets the size of the Data Store buffer	
PM:DS:SIZE?	NONE	Return the sizes of the Data Store buffer	
PM:DS:INTerval	1	Set data store interval.	SFREQ
PM:DS:INTerval?	NONE	Return data store interval.	SFREQ?
PM:DS:SAVEBUFER	1	Saves the data store buffer to USB flash disk	
PM:DS:SIZE	1	Set data store max size.	DSSIZE_n
PM:DS:SIZE?	NONE	Return data store max size.	DSSIZE_n?
PM:DS:UNITs?	NONE	Return data store units.	DSUNITS_n?
PM:FILTTer	1	Selects the filtering operation: no filtering, analog filter, digital filter, or analog and digital.	FILTER_n

Name	Number of Parameters	Function	Legacy Commands
PM:FILTer?	NONE	Returns the filtering operation: no filtering, analog filter, digital filter, or analog and digital.	FILTER_n?
PM:Lambda	1	Sets the wavelength for use when calculating power.	LAMBDA_n
PM:Lambda?	NONE	Gets the selected wavelength in nanometers.	LAMBDA_n?
PM:MAX:Lambda?	NONE	Returns the longest calibrated wavelength in nanometers.	
PM:MAX:Power?	NONE	Returns current range's maximum readable power.	
PM:MIN:Lambda?	NONE	Returns the shortest calibrated wavelength in nanometers.	
PM:MODE	1		MODE_n
PM:MODE?	NONE	Returns the currently selected acquisition mode.	MODE_n?
PM:Power?	NONE	Returns the power in the selected units.	R_n?
PM:PWS?	NONE	Returns the power with status.	RWS_n?
PM:RANge	1	Selects the gain stage when making readings with the detector head within a range from 0 to 5 (with zero being the highest).	RANGE_n
PM:RANge?	NONE	Returns an integer indicating the current range.	RANGE_n?
PM:RESPosivity?	NONE	Gets the responsivity currently used for making power calculations.	RESP_n?
PM:RUN	1	Disables or enables the acquisition of data.	RUN_n, STOP_n
PM:RUN?	NONE	Returns the present acquisition mode.	RUN_n?, STOP_n
PM:STAT:MAX?	NONE	Return statistics buffer maximum value.	STMAX_n?
PM:STAT:MEAN?	NONE	Return statistics buffer mean value.	STMEAN_n?
PM:STAT:MIN?	NONE	Return statistics buffer minimum value.	STMIN_n?
PM:STAT:MAXMIN?	NONE	Return statistics buffer maximum-minimum value.	STMXMN_n?
PM:STAT:SDEViation?	NONE	Return statistics buffer standard deviation value.	STSDEV_n?
PM:Temp?	NONE	Returns the 918 detector's temperature in degrees Celsius.	
PM:TRIG:EXTeRnal	1	Set external trigger enable.	EXT
PM:TRIG:EXTeRnal?	NONE	Return external trigger enable.	EXT?
PM:TRIG:EDGE	1	Set external trigger edge select.	EXTEDGE
PM:TRIG:EDGE?	NONE	Return external trigger edge select.	EXTEDGE
PM:TRIG:HOLDOff	1	Set external trigger holdoff time.	EXTHOLDOFF
PM:TRIG:HOLDOff?	NONE	Return external trigger holdoff time.	EXTHOLDOFF?
PM:UNITs	1	Selects the units for readings.	UNITS_n
PM:UNITs?	NONE	Returns an integer indicating the selected units.	UNITS_n?
PM:REF:VALue	1	Sets the user reference value for use in relative or dB readings.	USRREF_n
PM:REF:VALue?	NONE	Returns the user reference value.	USRREF_n?
PM:REF:STOre	NONE	Sets the user reference value for use in relative or dB readings as the present reading.	STOREF_n
PM:ZEROSTOre	NONE	Sets the zeroing value with the present reading.	STOZERO_n
PM:ZEROVALue	1	Sets the zeroing value.	ZEROVAL_n
PM:ZEROVALue?	NONE	Gets the zeroing value.	ZEROVAL_n?
TEC:T?	NONE	Return TEC measured temperature.	

Table 8 Legacy Commands Reference

15 **Appendix D – Sample Programs**

15.1 Programming Samples

The CD will install some simple programming samples to get computer interfacing started. These are minimal samples and provided only for reference.

15.2 LabVIEW

LabVIEW programming samples separated in folders based on version of LabVIEW compiled with. SampleQuery.vi demonstrates the use of the drivers.

15.3 Microsoft® Visual Basic

A zip file in the application folder contains a simple Visual Basic project for communicating with the meter.

15.4 Microsoft Visual C++

A zip file in the application folder contains a simple Visual C++ project for communicating with the meter.

15.5 Microsoft .NET

A zip file in the application folder contains a simple .NET project for communicating with the meter.

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Appendix E – Disassembly Instructions

NOTE

These disassembly instructions are intended only for recycling at the end of the product lifetime.
For troubleshooting or servicing, users should contact the local Newport Corporation representative. There are no user serviceable parts inside the equipment.
Attempting to self-service the unit will void the warranty.

16.1 Disassembly instructions

Figure 66 shows an exploded version of the 1918-R.

For recycling purposes only, the disassembly steps are as follows:

1. Make sure the unit power cord is removed.
2. Remove the battery.
3. Remove any other cables: detectors, Analog Output monitor cables, USB, and ground cables.
4. Remove all 4 screws on the rear panel.
5. Remove the rear panel.
6. Remove the screws and posts of each assembly and unplug each assembly from the unit.

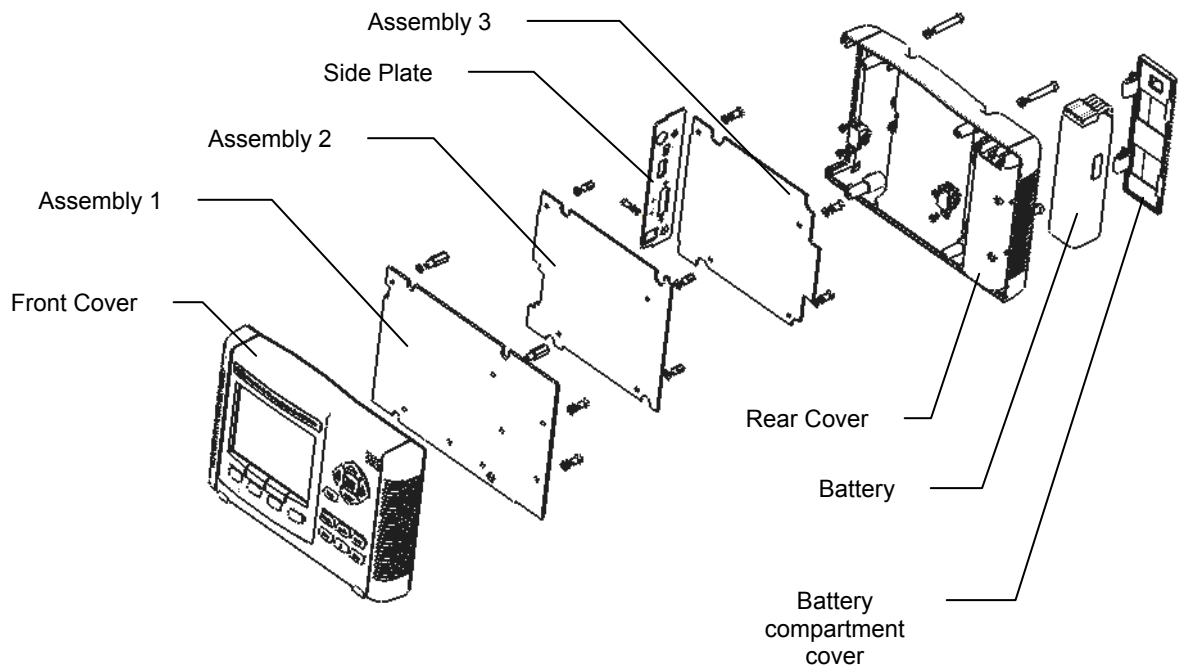


Figure 66 Disassembled 1918-R Optical Meter

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